When Does Graph Structure Help in Multi-Agent Reinforcement Learning? A Controlled Boundary

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Abstract

Graph neural networks are widely used to add coordination structure to cooperative multi-agent reinforcement learning (MARL), but gains are rarely separated from the capacity they add. We contribute a controlled-attribution method that does. On CoordGrid, a cooperative gridworld whose coordination graph is itself the experimental variable, we fix the encoder, mixer, optimiser, and parameter count and add two isolators: a byte-identical no-graph control (structure vs. capacity) and a density-matched *wrong*-graph control (the right edges vs. merely more edges). This reconciles a tension: graph-MARL methods such as DGN and G2ANet report gains in partially observable settings, rarely against capacity-matched controls; the discriminator between those positives and full-information negatives is the task's *information structure*, not the algorithm. When an agent must act on private information held by one specific, unobserved partner, a graph whose edges match the task's coordination structure decisively outperforms *all-to-all* communication—which carries the partner's signal but must select it—the wrong graph, and the no-graph control, at matched parameters and observations (Cohen's $d \approx 5$, complete seed-level separation over 12 seeds, Holm-corrected). Learned attention recovers the signal only on the right graph, even given 5× the training budget, where the gap widens ($d{\approx}5 \to 12$) rather than closes. The advantage holds across team sizes and observability, and replicates out of harness in a non-spatial referential game (matched graph ≈96% of maximum, rivals far below). When coordination is instead incidental or convention-reducible, the same architecture confers *no benefit* over the parameter-matched baseline; the commonly used *unstabilised* operator is actively harmful (a penalty that persists with training, grows with team size, and recurs on MPE `simple_spread` and, directionally, Level-Based Foraging)—but stabilisation removes this optimisation pathology, recovering the graph to parity. The rule: route along task-relevant edges, not all-to-all—and only when the coordination topology is known or reliably estimable. These are controlled, constructed tasks; replication on standard benchmarks with private per-agent goals is future work. Code, configurations, and data accompany the submission.

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) underlies progress on team coordination problems from traffic signal control to multi-robot manipulation. Value-decomposition methods, in which a joint team Q-value Q_{tot} is factorised into per-agent utilities Q_i for centralised training and decentralised execution, have emerged as a workhorse: VDN (Sunehag et al., 2018) factorises Q_{tot} as a sum, QMIX(Rashid et al., 2018; 2020) as a monotonic mixture, and several recent methods extend the mixer with graph neural networks (GNNs) over an externally specified or learned coordination graph (Jiang et al., 2020; Liu et al., 2020; Böhmer et al., 2020; Naderializadeh et al., 2020).

The intuition for graph-based MARL is straightforward: a model whose architecture mirrors the agents' coordination *structure* should generalise better and converge faster. Yet the conditions under which this holds are under-tested. Three specific gaps motivate the present study:

1. **Graph regime is rarely varied.** Most empirical comparisons fix one environment with one (often implicit) coordination graph, leaving open whether observed gains transfer to other graphs: sparse vs. dense, structured vs. random.
2. **Capacity is rarely controlled.** Adding a GNN adds parameters and depth as well as relational structure. Without a controlled comparison it is unclear whether the win comes from the graph or from the extra capacity.
3. **Receptive-field alignment is rarely studied.** A GNN of depth L propagates information across paths of length up to L. It is intuitive that L should grow with the graph's diameter. The general depth/long-range trade-off is well studied for GNNs (over-squashing on long-range tasks (Alon & Yahav, 2021), over-smoothing with depth (Li et al., 2018)), but we found no empirical study of receptive-field-versus-diameter alignment for a GNN inserted into a value-factorisation mixer (CTDE), specifically in the cooperative MARL setting.

A reusable attribution method, and the tension it resolves. A graph module changes a model's architecture and its capacity at once, so a measured win cannot be assigned to the graph's *structure* without a control—which is why the intuition above is under-tested, and what our design targets. We make the coordination graph the experimental variable and hold the encoder, mixer, optimiser, and parameter count fixed, then read the graph's contribution off two matched controls: a byte-identical *no-graph* control that isolates structure from capacity, and a density-matched *wrong-graph* control that isolates the right edges from merely more edges. These two controls are the paper's reusable methodological contribution, and we apply the same design at both endpoints below. They also resolve a standing tension in the literature: DGN and G2ANet report gains in *partially observable* settings but are rarely tested against capacity-matched controls, whereas controlled negatives arise in full-information regimes—two sides of one boundary whose discriminator, we argue, is the task's information structure, not the algorithm.

A two-endpoint answer. The title question is not a yes/no, and we answer it by locating *both* endpoints of a boundary under one controlled harness (Figure 1). At one endpoint—tasks that are full-information or *convention-reducible*, by which we mean solvable by a fixed shared convention the agents can adopt without exchanging private state (e.g. CoordGrid's edge-reward task, where meeting at a pre-agreed rendezvous solves coordination with no communication at all)—a graph-conditioned mixer confers no benefit over a parameter-matched no-graph control (and, in its common unstabilised form, is actively harmful, an optimisation pathology that stabilisation removes, Appendix H), with a capacity-matched control pinning the effect on the graph aggregation rather than on added parameters. At the other endpoint we *construct* the regime the first leaves unexplored: a task in which an agent must act on private information held by one specific partner it cannot observe, so the coordination graph is the only path for that information. There, message passing helps—but only when the communication topology matches the task's coordination edges; pointing the same channel at the wrong neighbours leaves it at the no-communication floor, and broadcasting to everyone does at most weak, task-incidental routing far below the task-matched graph. The two endpoints *coexist*: the positive result completes the negative one into a single "when does it help" statement rather than overturning it.

Contributions. This paper makes the following contributions.

1. We introduce CoordGrid (Figure 2A), a cooperative gridworld whose coordination graph is a tunable knob (ring, line, complete graph, lattice, Erdős–Rényi). The reward is defined edge-wise over this graph, so the graph is literally the coordination structure rather than a heuristic over a fixed task.
2. We implement IQL, VDN, QMIX, and GNN-QMIX in a single harness with shared encoder, mixer, optimiser, and exploration schedule (Figure 2B), so the only difference between GNN-QMIX and QMIX is the GNN. To isolate the graph from the capacity it adds (Gap 2) we include MLP-QMIX, a parameter-matched no-graph control (byte-identical parameter count, no neighbour aggregation); and GAT-QMIX, a graph-*attention* variant that tests whether the result is specific to the fixed GCN.
3. We pre-register three falsifiable hypotheses (Section 4) and test them with a pilot, a graph-regime sweep, and a depth × diameter ablation (Sections 6.1–6.4).

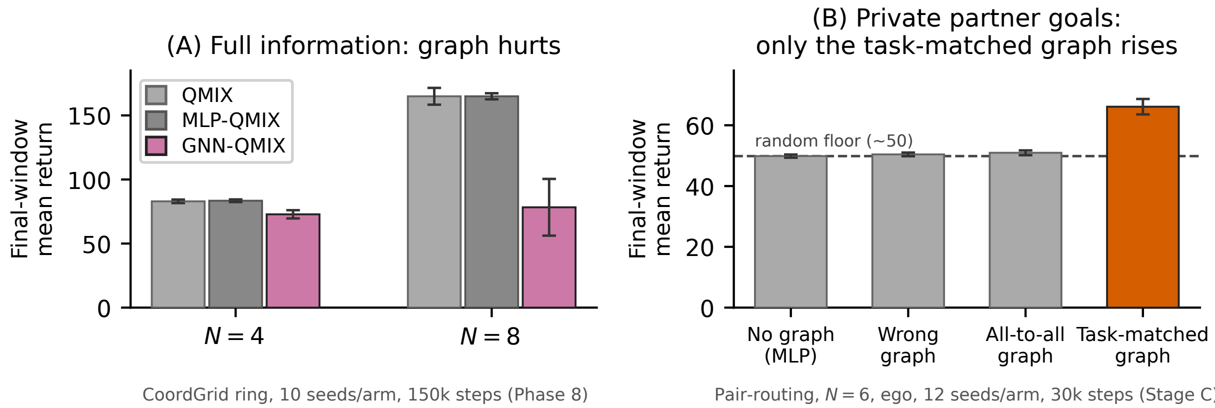


Figure 1: **Both boundary endpoints at a glance. (A) Negative endpoint:** on full-information CoordGrid (ring graph) the graph buys nothing—and the standard *unstabilised* GNN-QMIX (capacity-matched, +graph) shown here falls below both QMIX (no graph) and the parameter-matched MLP-QMIX (no graph) control at N=4 and N=8 (10 seeds, 150k steps; 95% CI). That deficit is an optimisation pathology: a *stabilised* GCN recovers to parity with the controls (Appendix H), still without any benefit. **(B) Positive endpoint:** on the pair-routing task, where each agent must act on a partner's privately held goal, only the task-matched graph rises well above the random floor (≈50); a density-matched wrong graph and the all-to-all graph stay at or within a point of the floor (12 seeds, 30k steps; 95% CI). The *same stabilised* graph-conditioned architecture is thus inert under full information (parity, panel A via Appendix H) yet a large asset when the task requires routing information the receiver cannot otherwise observe (panel B)—the discriminator is the task's information structure, not the architecture.

4. **The negative endpoint: in full-information / convention-reducible settings the graph confers no benefit—and its common unstabilised form is actively harmful for an optimisation reason, not because of added capacity.** The standard GNN-QMIX is bested by QMIX in both mean return and seed-level variance, even on an edge-defined-reward task whose observations (under the non-ego obs mode used there) already expose neighbour displacements, so message passing is at best redundant; the non-trivial finding is the capacity-matched *identification* of the cause (§6.5: the no-graph MLP-QMIX matches QMIX while GNN-QMIX falls far below both). The unstabilised-graph deficit survives a 5× longer training budget at n=10 seeds, grows monotonically with team size out to N=16 (distinct from the positive structure effect of contribution 5, which is significant at every N but non-monotone), reappears under the capacity decomposition on MPE `simple_spread` (where the naive GNN-QMIX-vs-QMIX comparison is null; §6.3), and persists under graph attention—but stabilising the GCN recovers it to parity with the controls (Appendix H), so what survives for the stabilised operator is the weaker, robust *no-benefit* claim.

5. **The positive endpoint: when an agent needs private information held by a specific other, the task-matched graph wins decisively—and the win is the structure** (§7). On a pre-registered pair-routing task in which each agent must reach a partner's *private* goal, the task-matched graph beats all-to-all communication—which carries the partner's information but must select it—at byte-identical parameters and observations ($d \approx 5$ at degree-1; Welch/Holm and Mann–Whitney $p < 10^{-4}$), while a density-matched wrong graph floors alongside the no-graph control: the right edges, not more edges. Expressiveness does not substitute for structure (stabilisation-matched learned attention stays near the floor over all-to-all even at 5× the training budget, while an exploratory multi-head arm gains +5.4 *given* the right graph), and the effect is significant at every N=4–10 (non-monotone), robust to observability, survives attenuated at degree-2, and replicates out of harness in a non-spatial token-matching game.

6. The depth–diameter alignment heuristic (H3) is falsified in favour of a sharper practitioner rule (§6.4): a hard over-depth cliff means GCN depth should not exceed two in small-team CTDE settings.

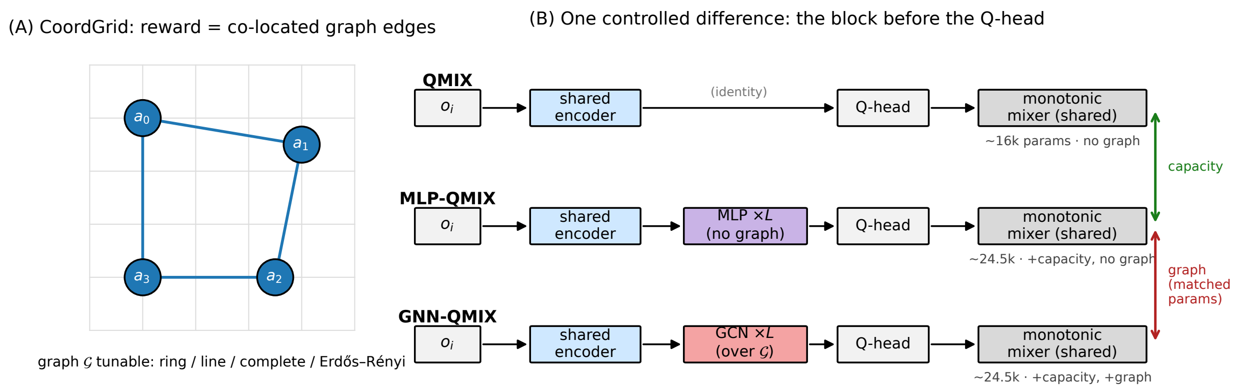


Figure 2: **Setup and controlled comparison.** (A) CoordGrid: agents act on a grid wired by a *tunable* coordination graph $\mathcal{G}$; the reward is the number of graph edges whose endpoints are co-located, so the graph *is* the coordination structure. (B) All algorithms share an identical per-agent encoder and monotonic mixer and differ only in the block inserted before the Q-head. QMIX inserts nothing; GNN-QMIX inserts an L-layer GCN over $\mathcal{G}$; MLP-QMIX inserts an L-layer MLP of *identical parameter count* but no graph. GNN-QMIX vs. MLP-QMIX isolates the *graph* (matched parameters); MLP-QMIX vs. QMIX isolates the *capacity*.

2 Background and Notation

Decentralised partially observable Markov decision process (Dec-POMDP). A cooperative MARL task is a tuple $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\mathcal{O}_i\}, \{O_i\}, \gamma \rangle$, where $\mathcal{N} = \{1, \ldots, N\}$ are agents, $\mathcal{S}$ the global state, $\mathcal{A}_i$ the discrete action set of agent i, $P(s' \mid s, \mathbf{a})$ the transition kernel, $R(s, \mathbf{a})$ a shared scalar reward, and $O_i(s) \in \mathcal{O}_i$ the local observation function. Agents share the discount γ and select actions from $\mathcal{O}_i$; centralised training (with access to s) is permitted while execution is decentralised (Oliehoek & Amato, 2016).

Centralised training with decentralised execution (CTDE) and value factorisation. Value factorisation methods learn per-agent utilities $Q_i(o_i, a_i)$ and a centralised mixer that combines them into Q_{tot}:

$$Q_{\text{tot}}(s, \mathbf{a}) = f_{\text{mix}}\big(Q_i(o_i, a_i)\,;\, s\big),$$

trained against a Bellman target on the team reward. The decentralised greedy policy $\pi_i(o_i) = \arg\max_{a_i} Q_i(o_i, a_i)$ recovers the joint greedy action when f_{mix} satisfies the Individual-Global-Max (IGM) condition (Son et al., 2019).

Coordination graph. We define a coordination graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ over the agents. CoordGrid (Section 5.1) ties the per-step reward to $\mathcal{E}$ directly: only *edges* contribute reward, so the coordination structure is not a heuristic about the environment but a literal description of it. GNN-QMIX takes $\mathcal{G}$ as side information; IQL/VDN/QMIX do not.

3 Related Work

Value-decomposition MARL. IQL (Tan, 1993) ignores agent interaction and trains independent Q-learners on the shared reward, which is biased but a useful no-coordination baseline. VDN(Sunehag et al., 2018) sums per-agent utilities. QMIX(Rashid et al., 2018; 2020) learns a monotonic state-conditioned mixer via a hypernetwork, enforcing $\partial Q_{\text{tot}}/\partial Q_i \geq 0$ so decentralised greedy execution is consistent with the centralised argmax. QTRAN (Son et al., 2019) and MAVEN (Mahajan et al., 2019) relax or extend the monotonicity assumption; QPLEX (Wang et al., 2021) reformulates it through a duplex dueling architecture. Beyond value-factorisation, MAPPO (Yu et al., 2022) shows that vanilla PPO with centralised critics is a surprisingly strong cooperative baseline; we restrict scope to value-decomposition methods to keep the architectural contrast with GNN-QMIX clean.

Learned and pre-specified inter-agent communication. Pre-dating the graph-MARL literature, RIAL/DIAL (Foerster et al., 2016) and CommNet (Sukhbaatar et al., 2016) learn pairwise or all-to-all communication channels between agents. These methods can be viewed as implicitly learning a (typically dense) coordination graph rather than taking one as input. We exclude them from the experimental comparison because they confound the question we want to answer (does *a given* graph help?) with the orthogonal question (can the agents learn a useful graph?).

Graph-based MARL. DGN (Jiang et al., 2020) applies graph convolution over a k-NN agent graph, evaluated on both discrete and continuous cooperative tasks. G2ANet (Liu et al., 2020) uses attention to abstract a game graph. Deep Coordination Graphs (Böhmer et al., 2020) factor Q_{tot} over a hyperedge structure. GraphMIX (Naderializadeh et al., 2020) interleaves a GNN with a QMIX-style monotonic mixer, closest in spirit to the GNN-QMIX baseline we evaluate here. Our GNN-QMIX is a deliberately *minimal* instantiation of this GraphMIX-style "GNN + monotonic mixer" family rather than a re-implementation of GraphMIX's full pipeline: it takes the coordination graph as an *external input* (rather than learning it), inserts a fixed-GCN encoder *before* an otherwise-unchanged QMIX mixer, and leaves the mixer untouched—so the GCN is the only structural variable, which is what makes the wrong-graph/all-to-all contrasts clean isolators (and is not intended as a strawman: it is chosen to isolate the GCN, not to weaken the baseline).

It is useful to separate these methods along a *fixed-vs-learned-graph* axis, because that axis is exactly what our controls exploit. At one end, DCG and our GNN-QMIX/GraphMIX take the interaction graph as an explicit input. DGN sits between: its graph is a *fixed* k-NN-by-proximity rule, recomputed as agents move but not learned. At the other end the graph is *inferred*—G2ANet's learned attention graph, DICG (Li et al., 2021), which infers a dynamic coordination graph for a GNN, and the context-aware sparse coordination graphs of CASEC (Wang et al., 2022) and Non-Linear Coordination Graphs (Kang et al., 2022). Of these, CASEC is the closest qualitative precedent for our headline: it argues and shows that *sparse* coordination graphs outperform dense/static ones. We credit that "sparse beats complete" is already established; our delta is the controlled *attribution*. CASEC (and the learned-graph line generally) *learns* the sparse topology and measures task return, so its gain entangles sparsity, learned selection, and capacity. We instead *fix* the topology and contrast a task-matched graph against a *density-matched wrong* graph and all-to-all at byte-identical parameters and observations—the density-matched wrong graph is what isolates the effect to *which* edges rather than to sparsity or to the act of learning a graph. Even DCG, the closest fixed-graph relative, defines payoffs over *given* hyperedges but never runs a wrong-graph control, so it cannot separate topology-correctness from the mere presence of a graph. More generally these methods combine three design choices (a graph, a message-passing mechanism, and a value mixer) in ways that make it difficult to attribute observed gains to any single component; our study isolates the GNN encoder by inserting it into an otherwise-identical QMIX architecture.

Recent coordination-graph and relational MARL (2023–2026). The learned-topology line has continued past CASEC. GACG (Duan et al., 2024a) infers *group-level* (higher-order) coordination edges, and LTS-CG (Duan et al., 2024b) learns a latent *temporally sparse* graph; both report gains from sparser learned topology and so inherit the same attribution gap we target—sparsity, learned selection, and capacity move together, so a return improvement cannot be credited to topology alone. ExpoComm (Li et al., 2025) reaches a conclusion adjacent to our headline from the opposite direction: small-diameter *exponential* communication graphs scale better than all-to-all broadcast, reinforcing that *which* edges exist matters more than how many—though it designs the topology for scalability rather than isolating topology-correctness against a wrong-graph control. Closest in spirit, Utke et al. (2025) ask directly when *relational* state abstraction helps cooperative agents and find it improves sample efficiency on spatially structured tasks; they use a learned relational critic and measure return, however, and run neither a density-matched wrong-graph nor a parameter-matched no-graph control. To our knowledge, no published work—through 2026—isolates the contribution of graph *structure* as we do: fixing the topology and contrasting a task-matched graph against a density-matched wrong graph and all-to-all at byte-identical parameters and observations. That is the gap this study fills.

Communication under partial observability. A parallel line of work motivates inter-agent message passing precisely by partial observability: when an agent cannot see what it needs, a learned channel can carry it. CommNet (Sukhbaatar et al., 2016) and DIAL (Foerster et al., 2016) learn such channels; DGN (Jiang et al., 2020) and G2ANet (Liu et al., 2020) make the channel graph-structured (a fixed k-NN graph and a learned attention graph, respectively). A closely related line argues specifically that *targeted* communication beats one-size-fits-all broadcast: TarMAC (Das et al., 2019) learns signature-based targeted messages, IC3Net (Singh et al., 2019) learns *when* to communicate, and ATOC (Jiang & Lu, 2018) learns an attentional communication group. This qualitatively anticipates our finding that all-to-all is worse than the right targeted edges—but those works *learn* the targeting and run no density-matched wrong-graph or parameter-matched no-graph control. We revisit this in §7.6: a learned attention head over all-to-all does not recover the matched-graph advantage *within our training budget*, which is consistent with—not contradictory to—TarMAC/ATOC succeeding, since those methods invest substantial training in learning to target, and is not a claim that attention cannot learn the targeting in principle. Our delta is distinct and complementary: we *fix* the topology and contrast a task-matched graph against a density-matched wrong graph and all-to-all at byte-identical parameters, isolating the contribution of *structure* from that of the channel and of the learned routing—a control these works do not run. The positive endpoint we identify sits squarely in this regime—private information that no agent can observe locally—and is what reconciles it with our negative result: a coordination graph earns its keep exactly when the task makes some agent's information unavailable to the others, and not in the full-information settings where prior controlled comparisons (and ours) find it inert. Where these methods entangle "does *a* graph help?" with "can the agents *learn* a useful graph?", we hold the graph fixed and vary only its *topology* relative to the task, which isolates the contribution of structure from that of the channel.

Foundational GNNs. We use a textbook graph convolutional network (Kipf & Welling, 2017); Veličković et al. (2018) and Xu et al. (2019) characterise alternatives. Battaglia et al. (2018) surveys the broader graph-network family.

Evaluation methodology. Our protocol follows the recommendations of Henderson et al. (2018) and Agarwal et al. (2021): multiple seeds, confidence intervals, an explicit sample-efficiency metric, and Holm-corrected pairwise tests in place of $p < 0.05$ readings off a forest plot. We additionally report effect sizes (mean differences) so readers do not have to infer them from CI overlap. Controlled comparisons of value decomposition versus independent learning already exist: Papoudakis et al. (2021) benchmark tuned IQL/VDN/QMIX across MPE/LBF/SMAC and find independent learning *competitive* on the fully-observable MPE/LBF tasks while value decomposition wins where coordination is genuinely required—squarely the territory of our negative endpoint. They do not, however, include a graph-conditioned mixer or a parameter-matched no-graph control; our contribution is to isolate the graph *aggregation* itself and attribute the penalty to it.

Benchmarks. The Multi-agent Particle Environment (Lowe et al., 2017) and PettingZoo (Terry et al., 2021) provide canonical cooperative tasks; SMAC (Samvelyan et al., 2019) is a heavier StarCraft micromanagement benchmark. We use CoordGrid as the primary test bed because it lets us *set* the coordination graph rather than infer it, and report a restricted MPE replication in Section 6.3.

4 Hypotheses

We test three pre-registered hypotheses:

H1 (structure beats no-structure):

On a cooperative task whose reward is graph-edge-defined, GNN-QMIXachieves a higher final return and crosses a fixed performance threshold sooner than IQL, VDN, and QMIX when the underlying coordination graph is sparse and structured.

H2 (structure > matched-density random):
The advantage of GNN-QMIX over QMIX is larger on a structured graph (ring) than on an Erdős–Rényi graph of the same edge density. This rules out the explanation that GNN-QMIX wins simply by having more inputs.

H3 (depth–diameter inverted-U):
Final return as a function of GNN depth L peaks near $L \approx d$, where d is the graph diameter, and degrades when L is either substantially smaller (insufficient receptive field) or substantially larger (over-squashing, optimisation noise). *Falsifiability criterion (pre-registered):* for each diameter d tested, we declare a per-row match if (a) the mean depth-return curve is non-flat (relative range $(\max - \min)/\max \geq 0.10$), and (b) $|L^* - d| \leq 1$ where L^* is the empirical argmax depth. We declare the family-level H3 confirmed if at least $\lceil 2n/3 \rceil$ of the n tested diameters pass the per-row criterion.

5 Experimental Design

5.1 The CoordGrid environment

CoordGrid is a cooperative gridworld parameterised by the team size N, the grid side G, the episode length T, and a coordination graph $\mathcal{G}$. The grid is toroidal. Each agent occupies a cell and chooses one of five actions per step: stay, up, down, left, right. The reward at step t is

$$r_t \;=\; \sum_{(i,j)\in\mathcal{E}} \mathbf{1}\big[p_i^{(t)} = p_j^{(t)}\big],$$

where $p_i^{(t)}$ is agent i's grid position at t. Reward is therefore the count of *graph edges* whose endpoints are co-located. Agents not linked by an edge have no reason to coordinate spatially: the graph is literally the coordination structure. Each agent observes its own normalised position, the relative displacement to each of its graph neighbours (zero-padded to a fixed slot count), and an agent-id one-hot. The global state used by the mixer is the concatenation of all agents' normalised positions.

Tunable graph regimes. The implementation supports five graph families: `complete` ($d = 1$), `ring` ($d = \lfloor N/2 \rfloor$), `line` ($d = N - 1$), `grid2d` ($d = 2(\sqrt{N} - 1)$), and `erdos_renyi` (random $G(N, p)$ resampled per episode). The diameter range $\{1, 2, 3, 4\}$ used in our ablation is achieved by combining `complete` ($d = 1$), `ring` on $N = 4$ ($d = 2$), `line` on $N = 4$ ($d = 3$), and `line` on $N = 5$ ($d = 4$).

5.2 Algorithms

The five core algorithms below (IQL, VDN, QMIX, GNN-QMIX, and the MLP-QMIX control; a sixth, the graph-attention variant GAT-QMIX, is introduced for the robustness check in §6.5) share a per-agent Q-network, an MLP of the form $\mathcal{A}_i \mapsto \text{ReLU}(\text{Linear}_{64}) \to \text{ReLU}(\text{Linear}_{64}) \to \text{Linear}_{|\mathcal{A}_i|}$ with parameters shared across agents and an agent-id one-hot appended to the observation. They share a 5×10^{-4} Adam optimiser, $\gamma = 0.99$, replay capacity 5×10^4, batch size 32, target hard updates every 200 gradient steps, Huber loss with Double-DQN targets (van Hasselt et al., 2016), and gradient norm clipping at 10.

- IQL: per-agent TD on the shared reward, no mixer.
- VDN: $Q_{\text{tot}} = \sum_i Q_i(o_i, a_i)$.
- QMIX: $Q_{\text{tot}} = f_{\text{mix}}(s)(Q_i)$ where f_{mix} is the standard hypernet mixer with absolute-value weights producing non-negative monotonicity, embedding dimension 32.
- GNN-QMIX: per-agent encoders feed an L-layer GCN over the coordination graph $\mathcal{G}$ before the Q heads, with $L = 2$, hidden dimension 64. The mixer is *identical* to QMIX; the GCN is the only structural change.

- MLP-QMIX (parameter-matched no-graph control): identical to GNN-QMIX in every respect except that the L-layer GCN is replaced by an L-layer MLP of the same per-layer shape, i.e. the neighbour-aggregation step $\tilde{A}H$ is removed, leaving $\mathrm{ReLU}(HW)$. MLP-QMIX has a byte-identical parameter count to GNN-QMIX at matched (L, width), so the only difference between the two is whether information crosses the coordination graph.

Why this matters: isolating graph from capacity. The mixer, optimiser, exploration schedule and per-agent encoder *stem* are identical across QMIX, GNN-QMIX, and MLP-QMIX. GNN-QMIX inserts an L-layer GCN of width 64 between the encoder and the Q head, adding $L \cdot (64^2 + 64) \approx 4.2\,L$ thousand parameters that QMIX does not have (e.g. $\sim$ 8.3k at $L = 2$, against $\sim$ 16k for QMIX). A comparison of GNN-QMIX against QMIX alone therefore confounds two explanations: the relational inductive bias of the *graph*, or the extra *capacity*. MLP-QMIX removes the confound by construction: it carries exactly the same extra parameters but no graph. This yields a three-way decomposition (verified to identical parameter counts in Section 6.5): QMIX (no capacity, no graph), MLP-QMIX (capacity, no graph), GNN-QMIX (capacity and graph). The fully-identified test of "does the graph help?" is GNN-QMIX vs. MLP-QMIX at matched capacity; MLP-QMIX vs. QMIX separately measures whether the capacity alone matters. The depth-vs-diameter ablation (Section 6.4, H3) additionally isolates depth effects within GNN-QMIX.

The full shared training loop (CTDE, off-policy, Double-DQN; identical across algorithms modulo the IQL per-agent loss) is given as Algorithm 1 in Appendix A.

5.3 Evaluation protocol

Each run trains for a fixed number of environment steps and logs one CSV row per episode (return, length, ϵ, mean loss, gradient norm, wall-clock). We report two metrics:

1. **Final-window return**: mean return over the last 10% of episodes of a run. This is a coarse capacity measure. (The Stage C/D cells of Section 7 use the last-20% window pre-registered for them; the difference is disclosed and checked in that section's preamble.)
2. **Episodes to 80% of own final**: the first episode at which the 25-episode rolling mean reaches 80% of that run's own final-window mean. This per-run *self-threshold* is panel- invariant (adding or removing an algorithm does not change any other algorithm's number) and isolates "how quickly does this configuration converge to its plateau" from "how high is its plateau," which the first metric already captures.

Across-seed aggregates use the sample mean and the Student-t 95% confidence interval. Pairwise algorithm comparisons use Welch's two-sample t-test as the primary and Mann–Whitney U as a non-parametric robustness check, with Holm–Bonferroni step-down correction. We define the multiple-test families explicitly: within a single condition (fixed N, fixed graph) Holm corrects across all algorithm pairs—6 pairs in the four-algorithm phases (1–4) and 10 pairs in the five-algorithm phases (3 and 8, which add MLP-QMIX); Phase 8's decisive conclusions additionally use the separately declared three-contrast decisive family (§6.5); H1 (within-condition GNN-QMIX vs. QMIX) is one one-sided test per condition and is not corrected within H1; H2 (interaction across graph regimes) is a single one-sample t on the per-seed difference-of-differences $\Delta_{\mathrm{ring}} - \Delta_{\mathrm{ER}}$ against zero and is therefore uncorrected. Confidence intervals use the Student-t 95% interval across the 10 seeds run per condition; we report effect sizes (mean differences) alongside p-values so readers do not have to back-fit power from CI overlap. Number of seeds and step budget per phase are listed in Table 6; deviations from the originally pre-registered budget were forced by compute constraints and are noted explicitly.

6 Results

Roadmap. The negative-endpoint evidence arrives in five experiments, named by the identifiers of the project's registered experiment log: Phase 1 is the *pilot* (harness sanity check), Phase 2 the *scale-and-graph-regime sweep* (H1, H2), Phase 3 the *MPE external-validity replication*, Phase 4 the *depth–diameter ablation* (H3), and Phase 8 the *capacity control* that identifies the cause and rules out undertraining. The numbering is non-contiguous because the intermediate identifiers were infrastructure and analysis stages that produce no reported comparisons; we keep the log's numbering so that the paper, the pre-registration documents, and the artifact's `results/` directories agree.

6.1 Phase 1: Pilot

Setup. 4 agents, ring graph (diameter $d = 2$), 10 seeds, 5×10^4 environment steps per run. This phase is a harness sanity check: per `docs/PHASES.md`, the acceptance criterion is that the coordinated baselines (VDN, QMIX) clearly outperform IQL on a task with obvious coordination demand, and GNN-QMIX runs to completion.

Findings. Figure 15 and Table 14 show the per-algorithm learning curves and final-window returns. Three results:

1. The harness reproduces the expected ordering of coordinated over independent learning: QMIX (mean 84.02, 95% CI $[83.41, 84.62]$) > VDN (mean 66.02, CI $[62.57, 69.48]$) > IQL (mean 34.62, CI $[32.61, 36.64]$).[1] The IQL vs. VDN gap is Holm-significant ($p_{\text{Holm}} < 0.001$) and the IQL vs. QMIX gap strongly so ($p_{\text{Holm}} < 0.001$). The Phase-0 acceptance criterion is met.
2. GNN-QMIX is markedly *worse* than QMIX on this condition (mean 66.02 vs. 84.02), and the gap is not attributable to capacity overhead (GNN-QMIX has the extra GCN parameters; if anything, that would predict a small *advantage* due to added expressiveness). Instead, GNN-QMIX exhibits elevated seed-level variance: its 95% CI spans $[57.11, 74.92]$, more than an order of magnitude wider than QMIX's $[83.41, 84.62]$, with the weakest seeds stalling in the low-return regime of IQL (Figure 15).
3. This finding is the first sign that "does graph structure help?" is not a simple yes/no. On the easy case (small team, small diameter, abundant coordination signal), the simpler QMIX mixer is both more accurate and more stable. The remainder of the paper asks under what conditions the extra graph machinery in GNN-QMIX does pay for itself.

6.2 Phase 2: Scale and graph regime (H1, H2)

Setup. $N \in \{4, 8\} \times \{\text{ring}, \text{Erdős–Rényi with matched density}\} \times 4$ algorithms $\times$ 10 seeds at 3×10^4 env steps per run. We shorten the pre-registered 2×10^5 steps and drop the N=16 cell—the original $N \in \{4, 8, 16\} \times 5$-seed $\times 2 \times 10^5$ design needed ∼24 M env steps, beyond the single-CPU envelope we committed to (commodity CPU, no GPU)—while *raising* the seed count to 10. See Table 6.

Findings (H1, H2). H1 (GNN-QMIX > QMIX on a structured graph) is *not supported*: GNN-QMIX is below QMIX in every cell (one-sided $p = 1.0$; Table 17). H2 (structure > matched-density random, tested as the difference-of-differences $\Delta_{\text{ring}} - \Delta_{\text{ER}}$) is *mixed*: it is supported at N=4 (DiD +26.9, one-sided $p < 0.001$) but reverses and is non-significant at N=8 (DiD −10.0, $p = 0.70$; Table 18). We therefore declare H2 mixed—the structured-vs-random gap is not monotone in team size—rather than supported. Per-condition returns and both tests are in Appendix K. The shortened budget here cannot be accused of underpowering a positive H1 into a null: the capacity-controlled Phase 8 sweep at 5× this budget (150,000 steps; Section 6.5) reaches the *same* verdict with the gap *growing*, so longer training does not rescue H1.

6.3 Phase 3: External validity on MPE `simple_spread`

Every number so far comes from CoordGrid, where the coordination graph is the experimental instrument. To test whether the identified graph penalty is an artefact of that one synthetic environment, we replicate the entire capacity control on a canonical cooperative benchmark: PettingZoo `simple_spread`, where N agents cover N landmarks while avoiding collisions. We change *only* the environment: the five algorithms, the locked hyperparameters, the 150,000-step / 10-seed budget, and the Welch / Holm / Mann–Whitney protocol are identical to the capacity-control protocol of Section 6.5. Two scope limits are stated plainly rather than footnoted. First, this executes at 15% of the pre-registered $\geq 10^6$-step budget (Table 6), a compute-forced

[1] VDN and GNN-QMIX round to the same 66.02 at two decimals (66.023 and 66.015 respectively; Table 14); this is a coincidence of two independent runs, not a duplicated row—their CIs differ markedly ($[62.57, 69.48]$ vs. $[57.11, 74.92]$).

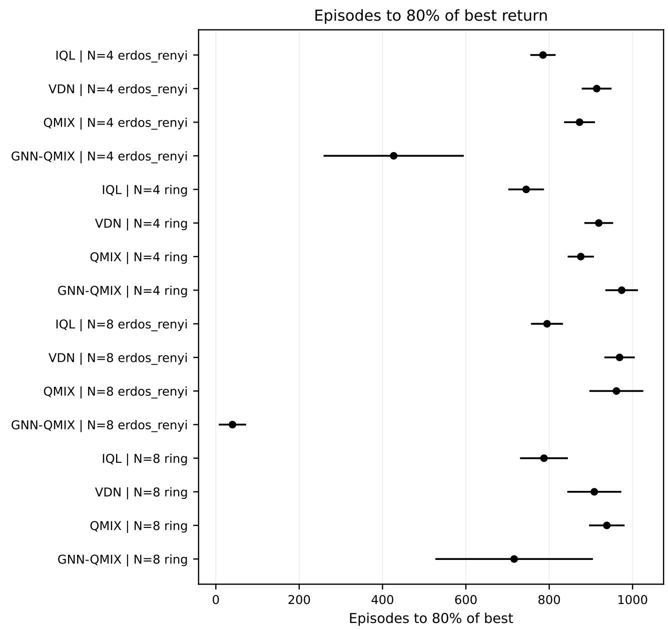
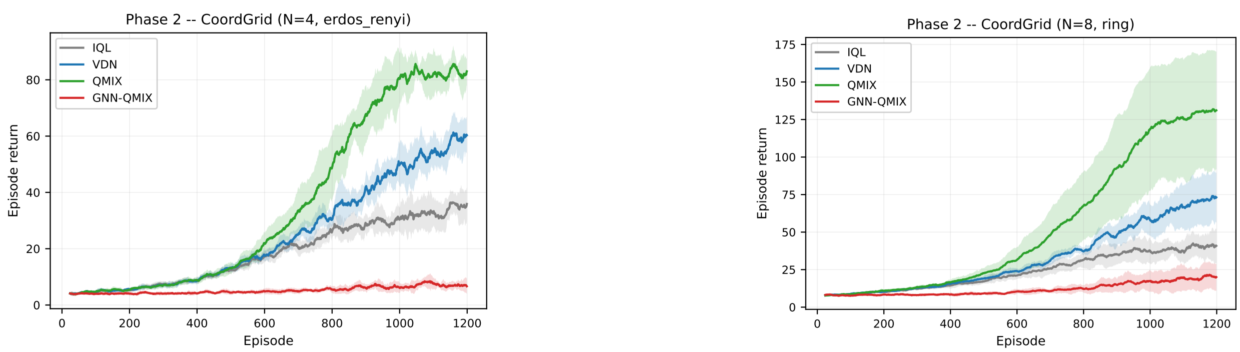


Figure 3: Phase 2 forest plot: episodes to 80% of each run's *own* final-window return, per algorithm × condition (the panel-invariant self-threshold metric of Section 5.3). Lower is better. Bars are 95% Student-t CIs across seeds. Per-condition returns and the H1/H2 hypothesis tests are in Appendix K (Tables 15, 17, 18).



(a) $N = 4$, Erdős–Rényi (matched density).

(b) $N = 8$, ring.

Figure 4: Phase 2 per-condition learning curves, the two most extreme cells of the scale/graph sweep. GNN-QMIX flat-lines near random-play return on the N=4 Erdős–Rényi cell, while QMIX climbs to ∼130 on the N=8 ring, its strongest Phase 2 cell (cross-seed mean of the rolling mean peaks at 131.9; the best individual seeds pass 150).

reduction. Second, at this budget the QMIX mixer family has not converged on `simple_spread`: the mixer-free IQL and VDN baselines lead it outright (Table 8). What this phase can and does test is therefore the *within-family decomposition at a fixed budget*—whether the graph helps or hurts relative to its own matched control—not converged final performance. `simple_spread` exposes no graph, so GNN-QMIX runs on a k-nearest-neighbour graph over agent positions (k=2, the DGN convention). At N=3 this graph is necessarily complete (each agent links to both others); N=6 is the genuinely sparse, larger-team test. The decomposition table (Table 7) is in Appendix E.

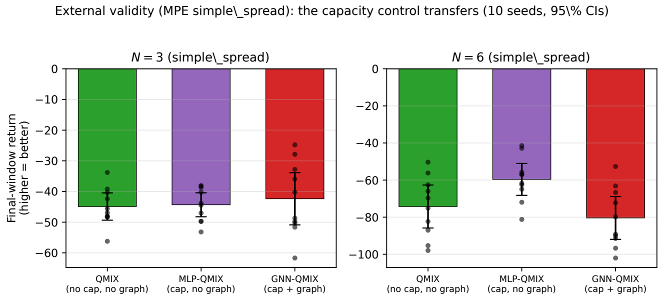


Figure 5: External validity on MPE `simple_spread` (10 seeds, 95% CIs; less negative is better). At N=6, MLP-QMIX (matched parameters, no graph) sits above QMIX while GNN-QMIX falls below both: the positive capacity effect and the negative graph effect the decomposition (Table 7) separates.

The graph penalty replicates, and the control is what reveals it. At N=3 all three decisive contrasts are null (graph effect +2.0, 95% CI $[-7.0, +10.9]$; capacity +0.6, CI $[-4.9, +6.1]$; both $p > 0.6$): on a small team with a complete graph we detect no effect in either direction—though the intervals are wide enough that this is a no-detection at n=10, not an equivalence claim. At N=6 the CoordGrid pattern reappears: the graph effect is −20.8 (Cohen's $d = -1.46$), significant under both Holm ($p = 0.014$) and Mann–Whitney ($p = 0.009$). The naive GNN-QMIX-vs-QMIX comparison is *not* significant here (net −6.2, $p = 0.41$): a single-comparison study would call the graph harmless. The control explains why: the extra parameters *appear to help* here (capacity +14.6, $d = +1.03$; Mann–Whitney $p = 0.045$, marginal after Holm—suggestive rather than confirmed, and we lean on it only as arithmetic), nearly cancelling the graph penalty ($+14.6 + -20.8 \approx -6.2$). Only MLP-QMIX separates the two, exposing a graph liability the headline comparison masks: on a standard benchmark, the confound this paper warns about.

What transfers, and what does not. The graph penalty grows with team size (null at N=3, $d = -1.46$ at N=6), mirroring CoordGrid. One effect is environment-specific: capacity is inert on CoordGrid but helpful on `simple_spread`, itself evidence that the controlled decomposition is necessary, since "GNN vs. no-GNN" means different things on different tasks. Destabilisation is milder here (GNN-QMIX/MLP-QMIX SD ratio ∼1.3–2.2× vs. roughly an order of magnitude on CoordGrid), so we do not claim it as general. Per-condition curves and the full five-algorithm table are in Appendix E.

6.4 Phase 4: Depth–diameter ablation (H3)

Setup. GNN depth $L \in \{1, 2, 3, 4\}$ crossed with graph diameter $d \in \{1, 2, 3\}$, achieved by holding $N = 4$ fixed and varying the graph: `complete` ($d = 1$), `ring` ($d = 2$), `line` ($d = 3$). Ten seeds, 2×10^4 env steps per run. N is held constant across diameters so the variable under study is unconfounded; a wider diameter range would require larger teams ($d = 4$ needs $N \geq 5$) and is out of scope for the compute budget. `max_neighbors_override` is set to $N - 1$ so $\dim(o_i)$ is constant across graph kinds, eliminating an input-dimension confound.

Predictions (pre-registered, from H3). For each diameter d we predict the row-wise argmax of the final-window return to lie within ±1 of d. Concretely: $d = 1 \Rightarrow L^* \in \{1, 2\}$; $d = 2 \Rightarrow L^* \in \{1, 2, 3\}$; $d = 3 \Rightarrow L^* \in \{2, 3, 4\}$. Curves must additionally be non-flat (relative range ≥ 10% of peak) for a row to count as a confirmation, to prevent noisy flat curves from passing by accident.

Findings. Figure 6 reveals a cleaner and more concerning pattern than H3 predicted:

1. **H3 is not confirmed at the family level** (1/3 diameters pass). Only the ring ($d = 2$) shows the predicted argmax-at-$L = d$ behaviour ($L^* = 2$, non-flat: pass). The complete graph ($d = 1$) is essentially flat in L (relative range 9%, below the pre-registered 10% non-flatness floor—which is exactly why it fails the per-row criterion) and the line ($d = 3$) shows the *opposite* of the prediction: $L^* = 1$, with a monotone decrease as L grows.

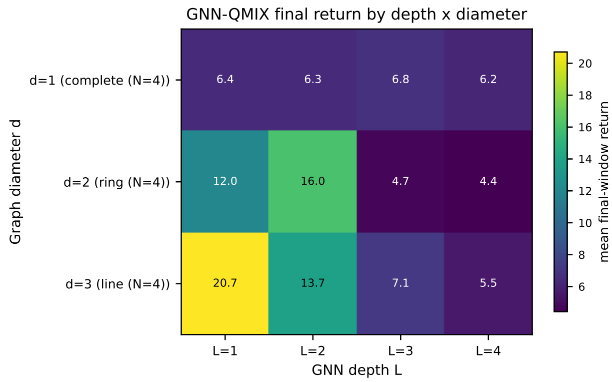


Figure 6: Phase 4: GNN-QMIX final-window mean return as a function of GNN depth L and coordination graph diameter d. Cells along the diagonal correspond to depth–diameter alignment ($L \approx d$), the H3 prediction. Outcome: H3 is *not* confirmed at the family level (1/3 diameters pass); on the structured diameters ($d = 2, 3$) the two rightmost columns ($L = 3, 4$) collapse—an over-depth cliff—while the complete graph ($d = 1$) is flat in L.

Table 1: H3 (depth–diameter alignment) per-diameter test, pre-registered criteria: (a) the depth curve must be non-flat (relative range $\geq 10\%$ of peak), AND (b) the argmax depth L^* must satisfy $|L^* - d| \leq 1$. Per-diameter passes: 1/3.

d	L^* (argmax)	Peak return	rel. range	pass
1	3	6.77	0.09	×
2	2	16.03	0.72	✓
3	1	20.71	0.73	×

2. Reading the heatmap column-wise rather than row-wise exposes a sharp *over-depth cliff* on the structured diameters: $L = 3$ and $L = 4$ collapse on $d \in \{2, 3\}$, while the complete graph ($d = 1$) stays flat in L. Mean returns at $L \in \{3, 4\}$ are 4.4–7.1 across the grid, while $L \in \{1, 2\}$ deliver 6.3–20.7. Whatever benefit additional message-passing layers contribute in principle, on the structured graphs it is dominated by an optimisation or over-squashing penalty that kicks in immediately past $L = 2$ in this regime.

3. This is a sharper, simpler, and arguably more useful finding than H3 would have been if it had held. The practitioner rule is not "set $L \approx d$," but rather "do not exceed $L = 2$ in a setting like ours, a small team with small replay buffer and seed budget, on tasks whose diameter is already small." Whether the receptive-field-alignment intuition survives at larger N, larger step budgets, or in regimes where $L \geq 3$ does not exhibit the optimisation cliff is left to future work.

6.5 Phase 8: Capacity control and the undertraining confound

The Phase 1–2 comparison of GNN-QMIX against QMIX leaves two explanations open, both flagged in our own limitations: the gap could be the graph, or it could be the ∼8.3k extra parameters the GCN adds; and the short (3×10^4-step) budget could have undertrained the larger model. Phase 8 closes both. We run the parameter-matched no-graph control MLP-QMIX (Section 5.2) alongside QMIX and GNN-QMIX on the decisive `ring` cells at $N \in \{4, 8\}$, for 10 seeds at 150,000 env steps each, 5× the Phase 2 budget and double the seed count Agarwal et al. (2021) recommend.

Setup. Five algorithms $\times\{N{=}4, N{=}8\}$ ring × 10 seeds at 150,000 steps. GNN-QMIX and MLP-QMIX use the locked $L{=}2$, width-64 encoder and are verified to have identical parameter counts, so the only difference is the neighbour-aggregation step. Metric is final-window return; we report Welch's t (Holm-corrected over the three decisive contrasts) and the non-parametric Mann–Whitney U, the latter robust to the variance

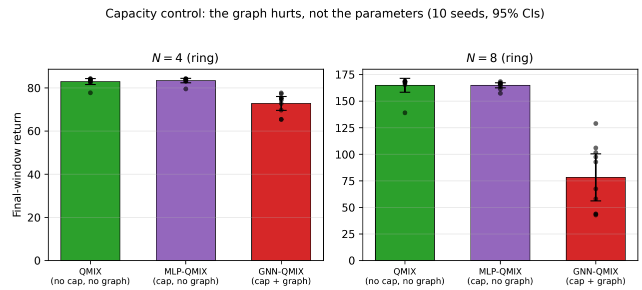


Figure 7: Capacity control on CoordGrid `ring`. MLP-QMIX (same parameter count as GNN-QMIX, no graph) tracks QMIX, while GNN-QMIX falls far below both at both team sizes: the penalty is the graph aggregation, not the added capacity. Means over 10 seeds; 95% Student-t CIs.

blow-up reported below. Figure 7 summarises the result (full per-cell numbers in Table 19, Appendix L); the findings below state the decisive values inline.

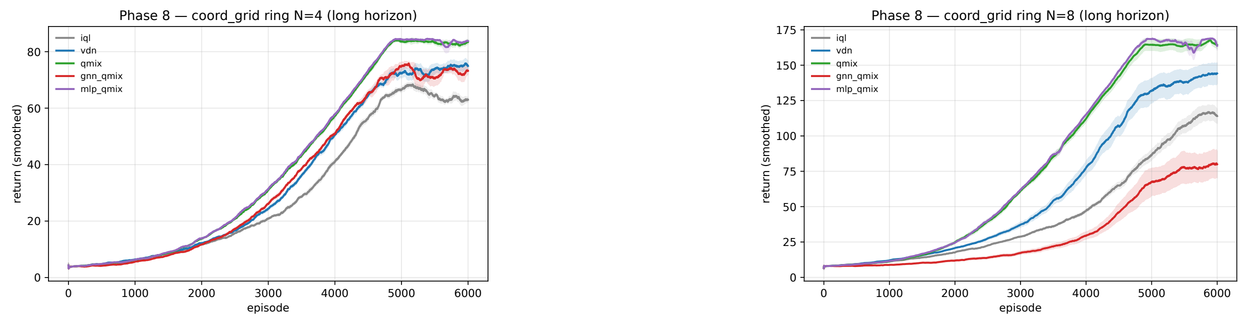
Finding 1: the penalty is the graph, not capacity, and it is significant. MLP-QMIX matches QMIX in mean at both scales (capacity gap 0.0 at $N = 8$; negligible at $N = 4$), so the extra parameters alone add nothing on average. For completeness: a rank test does detect a small MLP-QMIX-below-QMIX median deficit at N=8 (Mann–Whitney $p = 0.045$; $\sim$2 points, under 1.5% of QMIX's return, driven by equal means with unequal spreads)—an order of magnitude smaller than the graph effect, so we describe capacity as approximately inert rather than exactly zero. GNN-QMIX (the standard, unstabilised GCN), with identical parameters but the graph added, is far worse: the graph effect is -10.6 at $N = 4$ and -86.6 at $N = 8$ (Cohen's $d = -3.2$ and -3.9). Both decisive contrasts are significant under the non-parametric test (MWU $p = 0.0002$) and under Holm correction over the decisive family. The capacity control thus *identifies* the source: not the added parameters (MLP-QMIX matches QMIX) but the graph aggregation, and the deficit *grows* with team size—the opposite of the "graphs help coordination as teams grow" intuition. *An important scope note, established in Appendix H:* this large deficit is a property of the *unstabilised* GCN. Adding the same residual+LayerNorm stabilisation used at the positive endpoint recovers GNN-QMIX to parity with the no-graph controls (164.3 vs. 164.9 at N=8, $d \approx -0.1$; the $\sim 9\times$ variance inflation below also vanishes). So the harm is an optimisation pathology of the common operator, not an intrinsic cost of graph aggregation—but even the stabilised graph confers *no benefit* in this full-information regime (it matches the control, never beats it), which is the robust negative-endpoint conclusion.

Finding 2: undertraining is ruled out. At 150,000 steps (5× Phase 2), QMIX still dominates GNN-QMIX by roughly 2× in return at $N = 8$ (164.9 vs. 78.3). The gap is not a transient of insufficient training; it persists and widens at the larger budget.

Finding 3: the unstabilised graph destabilises learning. Beyond the lower mean, the plain GNN-QMIX is far less reliable across seeds: at $N = 8$ its seed standard deviation is 30.9, against 3.3 for the parameter-matched MLP-QMIX control and 9.1 for QMIX, roughly a 9× inflation. The plain graph does not merely fail to help on average; it injects seed-level instability that the parameter-matched no-graph control does not (Figure 8). This instability is exactly what the stabilised GCN removes (Appendix H: seed SD 7.0 vs. 30.9), which is why we read the destabilisation as an optimisation pathology of the operator rather than an inevitable consequence of message passing.

Finding 4: the pattern is not specific to the ring. Repeating the full capacity control on the higher-diameter `line` topology (10 seeds, 150,000 steps; Appendix G, Table 10) reproduces every effect: the graph penalty is -10.0 at $N = 4$ and -65.8 at $N = 8$ (both significant under Holm and Mann–Whitney, $p = 0.0002$), the capacity contrast is again null, and GNN-QMIX again carries by far the largest seed variance (SD 29.5 vs. 2.7 for MLP-QMIX at $N = 8$). The identified graph penalty thus holds across two distinct coordination-graph topologies.

Finding 5: the penalty survives a stronger GNN and grows with scale. Two robustness checks (Appendices I–J). *(i) GNN family.* The learned attention these methods use (Jiang et al., 2020; Liu et al.,



(a) $N = 4$ (ring).

(b) $N = 8$ (ring).

Figure 8: Phase 8 training curves (n=10 seeds, 150,000 steps). MLP-QMIX (the parameter-matched no-graph control) tracks QMIX at both scales, while GNN-QMIX is both lower and visibly more variable across seeds: the destabilisation of Finding 3.

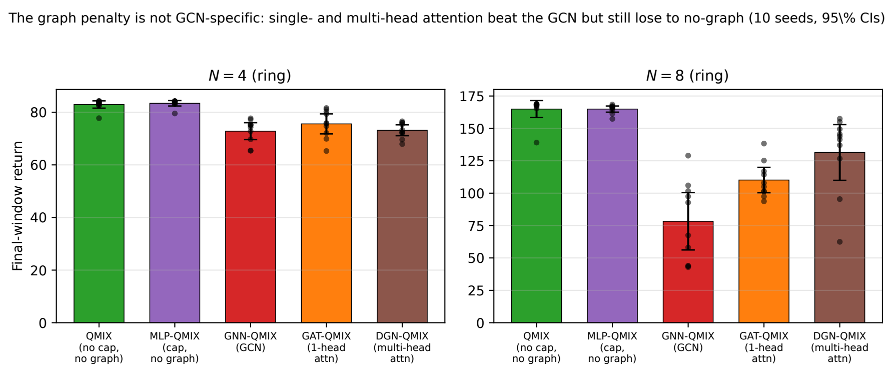


Figure 9: Second GNN family. Single-head (GAT-QMIX) and multi-head (DGN-QMIX, the DGN/G2ANet operator) graph attention both sit above GNN-QMIX (the fixed GCN), more expressive attention helps *within* graph methods, but still below the parameter-matched no-graph control MLP-QMIX at both team sizes on CoordGrid: no unstabilised graph operator recovers a benefit here (cf. Appendix H, where the *stabilised* GCN reaches parity but still does not exceed the control).

2020) does not rescue the graph. Beyond the GCN we run single-head (GAT-QMIX) and multi-head (DGN-QMIX, the DGN/G2ANet operator, our largest variant) attention; on CoordGrid *both* lose to the no-graph control at $N = 8$ ($d = -5.5$ and -1.6; Holm + MWU significant, likewise $N = 4$). More attention climbs *within* graph methods (DGN-QMIX > GAT-QMIX > GNN-QMIX at $N = 8$) yet never beats no-graph; on MPE it only narrows the gap (non-significant at N=6). Better attention helps the graph *approach*, not overtake, the control. *(ii) Scale.* Extending to $N \in \{12, 16\}$, the graph effect grows *monotonically*: -10.6, -86.6, -153.4, -267.4; at $N = 16$ GNN-QMIX reaches only ∼18% of QMIX's return. The "graphs help as the team grows" intuition is exactly inverted.

Operator scope of Findings 1–5. All five findings above use the standard *unstabilised* GCN (and its attention analogues). As Appendix H establishes, the large deficits, the variance inflation, and the growth with N are properties of that operator: stabilising the GCN (residual+LayerNorm) recovers it to parity with the no-graph controls on both the ring and line cells. The conclusion that survives for the stabilised operator, and that we carry forward as the negative endpoint, is the weaker but robust one—in this full-information regime the graph confers *no benefit over a parameter-matched no-graph control*. The "actively harmful / destabilising" language is scoped throughout to the common unstabilised operator.

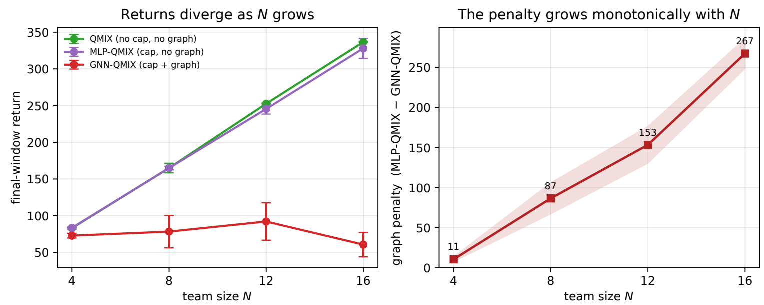


Figure 10: Scaling the graph penalty. *Left:* final-window return vs. team size: the graph-free curves (QMIX, MLP-QMIX) climb together while GNN-QMIX flatlines. *Right:* the graph penalty (MLP-QMIX − GNN-QMIX) grows monotonically with N, to ∼18% of QMIX's return at N=16.

7 When the graph earns its keep: a structure boundary

Everything so far is the *negative* endpoint: where any agent could infer what it needs without the graph, the graph aggregation is neutral to harmful. This section identifies the *positive* endpoint. We construct a task in which the coordination graph is the only path for task-relevant private information, and ask whether—channel, capacity, and observations held fixed—the graph's *structure* (which edges) earns a measurable, reproducible advantage. It does. We are precise about what was pre-registered: the hypotheses, metric, seeds, and decision rules for the core pair sweep, the team-size sweep, the degree-2 cell, and the attention-over-complete test were fixed before those runs (`docs/STAGE_C.md` in the artifact), following an exploratory 3-seed pilot; the full-observability cell (§7.4), the `dgn_true` arm, and the TokenMatch replication (§7.7) are post-hoc robustness/replication cells run under the same protocol and thresholds but not themselves pre-registered. We flag one limit on how independently this ordering can be checked: only the F1 attention robustness check (`results/phaseC_150k/PREREGISTRATION.md`) was committed to version control before any 150k-step result existed, so it alone is timestamp-verifiable as pre-committed; the earlier pre-registrations rest on the documented protocol rather than on commit-order evidence. All confirmatory cells use ≥ 12 seeds and the same Welch / Holm and Mann–Whitney tests of Section 5.3. One protocol difference is disclosed rather than hidden: the cells of this section use the final window pre-registered for them—the last *20%* of episodes—whereas Section 5.3 defines the last 10% for Phases 1–8; recomputing every contrast of this section at the 10% window leaves every conclusion unchanged with slightly *larger* gaps (e.g. `gnn_true` vs. `gnn_complete` +15.8, $d = 5.8$). The Holm family here is the registered `gnn_true`-vs-each-rival family—three contrasts in the core sweep (`gnn_true` vs. `gnn_wrong`/`gnn_complete`/`mlp`), six in the attention sweep (adding the `gat`/`dgn` arms)—not the six-algorithm-pairs family of Section 5.3, so the same contrast can carry slightly different p_{Holm} across the two sweeps (both correct per family).

7.1 A task where structure is load-bearing by construction

PairRouting is CoordGrid with a single change to the reward: every agent privately observes *its own* goal and is rewarded (dense toroidal proximity) for reaching its *partner's* goal, where a fixed perfect matching defines partners. Partners must therefore exchange goals across their edge; the information an agent needs is held by *one specific other agent*, so the communication graph must match the task structure for routing to be possible at all. We hold $\dim(o_i)$ constant across communication graphs by setting `max_neighbors_override`$= N-1$ (unit-tested), so no arm sees more inputs than another. The arms are the *same* GNN-QMIX architecture, in a *stabilised* ("repaired") variant: a residual connection and LayerNorm around each GCN layer (L=2, width 64). The repair exists because the plain GCN of Sections 6.1–6.5 does not train on this routing task at all (exploratory Stage-A runs); the negative-endpoint phases use the plain GCN, and Appendix H shows that giving that endpoint the *same* stabilisation recovers GNN-QMIX to parity with the no-graph controls—so the full-information "harm" is an optimisation pathology of the unstabilised operator, while the graph's *lack of benefit* there holds for the stabilised operator too. The positive-endpoint effect below therefore compares the stabilised operator against itself under different adjacencies, and cannot be an artefact of the plain-vs-stabilised difference. Only `env.adjacency()` differs across the graph arms: `gnn_true` (the matching), `gnn_wrong` (a *different* matching of identical 1-regular density), and `gnn_complete` (all-to-all, the most communication). `mlp` is the no-communication parameter-matched control—the same control as MLP-QMIX in Section 5.2; this section uses the short arm names of the pre-registration throughout. Because `gnn_wrong` and `gnn_complete` are byte-identical to `gnn_true` in architecture and parameter count—only the adjacency changes—the `gnn_true` vs. `gnn_complete` contrast isolates *structure* from channel, capacity, and observability simultaneously (the next paragraph explains why `gnn_complete`, and not `gnn_wrong`, is the load-bearing rival).

Which contrast carries the claim. On a routing task, an agent cannot route information it is barred from observing, so a graph beating the no-graph control is close to *a priori*. Two of our four arms carry zero partner information by construction and are therefore a-priori at the floor: `mlp` (no channel), and—we concede this to be the same information state—`gnn_wrong`, because a wrong matching links agent i to a non-partner, so the partner's private goal never reaches i (verified in `coord_grid.py`; the data agree, `gnn_wrong`≈`mlp`≈floor). We therefore treat both `gnn_true` vs. `mlp` and `gnn_true` vs. `gnn_wrong` as sanity checks confirming that mis-wired or absent channels cannot route, and rest the headline structure claim on the one genuinely non-a-priori contrast: `gnn_true` vs. `gnn_complete`. All-to-all is the only rival in which the partner's information *is* present in the inputs and the model must *select* it from the $N-1$ neighbours; that it nonetheless floors is the load-bearing evidence that the win is which edges carry messages, not whether the channel can in principle reach the partner. (For a fixed mean-aggregating GCN, averaging N node embeddings dilutes any single partner's goal to $\sim 1/N$, so this floor is partly structural for that aggregator class. Two pieces of evidence make the claim non-trivial despite this. First, §7.6 shows a *learned* attention head over all-to-all—which could in principle up-weight the partner rather than dilute it—also floors within budget. Second, §7.7 shows that the diluted all-to-all signal *is* usable when the task permits: there all-to-all rises +12 above the floor, demonstrating mean-pooling does not simply destroy the partner's signal—yet the matched graph still beats it by +33. So the all-to-all floor is not merely a mechanical dilution artefact; it reflects the absence of the right edges.)

How we found this task (failed constructions disclosed). PairRouting was not the first construction we tried, and the search matters for scope. An exploratory stage first tested whether restricting observability alone (making agents depend on the channel) or stabilising the GCN alone would surface a graph advantage on the convention-reducible tasks of Sections 6.1–6.5: neither did. A pre-registered goal-*broadcast* construction (one agent observes a shared secret goal that must relay over ring/line graphs; 10 seeds × 3 diameters × 4 arms) then failed its primary hypothesis: the matched graph's advantage at diameter 5 was +3.2 (Welch $p_{\text{Holm}} = 0.60$), so multi-hop *relay* did not produce a significant structure win for these operators. PairRouting moves the private information a single edge away (one specific partner), and that is where the advantage becomes large and clean; the degree-2 cell (§7.5) shows it surviving, attenuated, when the needed information is a two-neighbour *set* rather than a single partner. The positive rule of this paper is accordingly scoped to short-range, task-matched routing; making multi-hop relay work is an open problem these operators did not solve, not a suppressed null. The failed sweeps ship with the artifact (`results/phaseB/`).

7.2 Headline: it is the structure, not the channel

Setup. N=6, 3×3 grid, ego observations, 12 seeds, 30k steps. Two reference points anchor the scale, both measured on this task (script and CSV in the artifact: `scripts/phaseC_reference_points.py`, `results/phaseC/reference_points.csv`): the random / do-nothing return is 49.9 (≈50), and a full-information greedy *oracle*—every agent handed its partner's goal—reaches 148.7 (≈149, against the analytic ceiling of $N \cdot T = 150$). So the floor is literally the random policy, and the achievable routing headroom is $\sim$50 → $\sim$149.

More communication did not help; the right communication did. Table 2 foregrounds the controls; Figure 11 shows the same result at a glance. The no-graph control (`mlp`), the wrong graph (`gnn_wrong`), and *all-to-all* communication (`gnn_complete`) all sit at or within a point of the random floor—49.85, 50.42, and 50.94 respectively, more than 15 points below `gnn_true`. Only `gnn_true`, routing along the task's own edges, rises, to 66.10 [63.56, 68.65]. The decisive arm is `gnn_complete`: it is the only rival that actually *carries* the

Table 2: **The structure boundary** (PairRouting, N=6, ego, 12 seeds, 30k steps). *Top:* the wrong graph and the no-graph control sit at the random floor (≈50, measured); all-to-all sits one point above it; only the task-matched graph rises. *Bottom:* the three registered contrasts (same repaired GCN and parameter count; only the adjacency differs). The headline rests on `gnn_true` vs. `gnn_complete`, the only rival that carries the partner's information but must select it; `gnn_true` vs. `gnn_wrong` and `gnn_true` vs. `mlp` are a-priori sanity checks (neither carries partner information). d is Cohen's d; p_{Holm} is Welch's t with Holm correction over the registered three-contrast family (actual $\sim 3\times 10^{-8}$); Mann–Whitney is at its small-sample floor under complete separation. What the matched graph does and does not deliver of the $\sim 50 \to \sim 149$ headroom is discussed at the end of §7.2.

Arm	comm. graph	final return [95% CI]
`mlp`	none	49.85 [49.33, 50.38]
`gnn_wrong`	wrong matching (same density)	50.42 [49.81, 51.02]
`gnn_complete`	all-to-all (most comms)	50.94 [50.16, 51.73]
`gnn_true`	**the task matching**	**66.10** [63.56, 68.65]
Contrasts (only the adjacency differs; all hold capacity, architecture, obs. fixed):		
`gnn_true`−`gnn_complete` (decisive: partner info present, must be selected)	**+15.2**, $d = 5.1$, $p_{\mathrm{Holm}} < 10^{-4}$, MWU $< 10^{-4}$	
`gnn_true`−`gnn_wrong` (a-priori: wrong matching carries no partner info)	+15.7, $d = 5.4$, $p_{\mathrm{Holm}} < 10^{-4}$, MWU $< 10^{-4}$	
`gnn_true`−`mlp` (a-priori: no channel)	+16.3, $d = 5.6$, $p_{\mathrm{Holm}} < 10^{-4}$, MWU $< 10^{-4}$	

partner's information (the partner is one of the $N-1$ broadcast neighbours), yet on PairRouting it recovers almost none of it. The pre-registered comm-insufficiency check quantifies "almost none": `gnn_complete` ends +1.1 above the no-communication control (Cohen's d=1.0; Welch p=0.020, Mann–Whitney p=0.030)—a statistically detectable sliver of task-incidental routing amounting to ≈7% of the `gnn_true` gap. So the arm with the *most* communication is distinguishable from no communication only by a point, while the arm with the *right* communication clears both by more than fifteen. (The size of the sliver is task-dependent: in TokenMatch, §7.7, the same arm manages real partial routing, +12 over the floor, yet still falls −33 short of the matched graph. In neither task does broadcasting approach the right edges.) The advantage is which edges carry messages, not how many.

The decisive contrast. The headline rests on `gnn_true` vs. `gnn_complete` (all-to-all): +15.2 ($d = 5.1$, $p_{\mathrm{Holm}} < 10^{-4}$, Mann–Whitney $p < 10^{-4}$). This is the only contrast in which the partner's information *is* present in the inputs yet must be selected, so its near-floor result cannot be explained by the channel being unable to reach the partner. The other two contrasts are a-priori sanity checks, as established above: `gnn_true` vs. `mlp` (+16.3, $d = 5.6$; no channel) and `gnn_true` vs. `gnn_wrong` (+15.7, $d = 5.4$; a wrong matching carries no partner information by construction), both also at $p_{\mathrm{Holm}} < 10^{-4}$ and Mann–Whitney $p < 10^{-4}$. Distributional separation is complete across all three: the worst `gnn_true` seed (61.5) exceeds the best rival seed (52.6) over 12 seeds. (The Mann–Whitney p-values sit at their small-sample floor under complete separation: they indicate non-overlap, not additional effect magnitude.)

Parameter matching, precisely. `gnn_wrong` and `gnn_complete` are *exactly* parameter- and architecture-identical to `gnn_true` (only `adjacency()` differs), which is what carries the structure claim. The `mlp` control matches the *plain*-GCN parameter count; against the as-run *repaired* GCN it differs by the 256 LayerNorm parameters (< 2% of ∼14k), far too few to explain a d≈5 effect—and `gnn_wrong`/`gnn_complete` carry the same LayerNorm yet floor, so stabilisation is not the source of the win. **The graph helps; it does not solve the task:** `gnn_true` captures only ∼16% of the routing headroom (66 of 149 against a 50 floor).

7.3 Robustness across team size: significant everywhere, but non-monotone

We pre-registered that the advantage would *grow* with N (a dose-response). It does not (Figure 12, left). Sweeping $N \in \{4, 6, 8, 10\}$ (ego, 12 seeds, 30k), the task-matched advantage over the no-graph control is significant at *every* N under both tests and Holm correction—and the structure-isolating `gnn_true` vs. `gnn_wrong`/`gnn_complete` contrasts are likewise significant at every N—but the standardised and relative effect form an inverted-U and then *attenuate*:

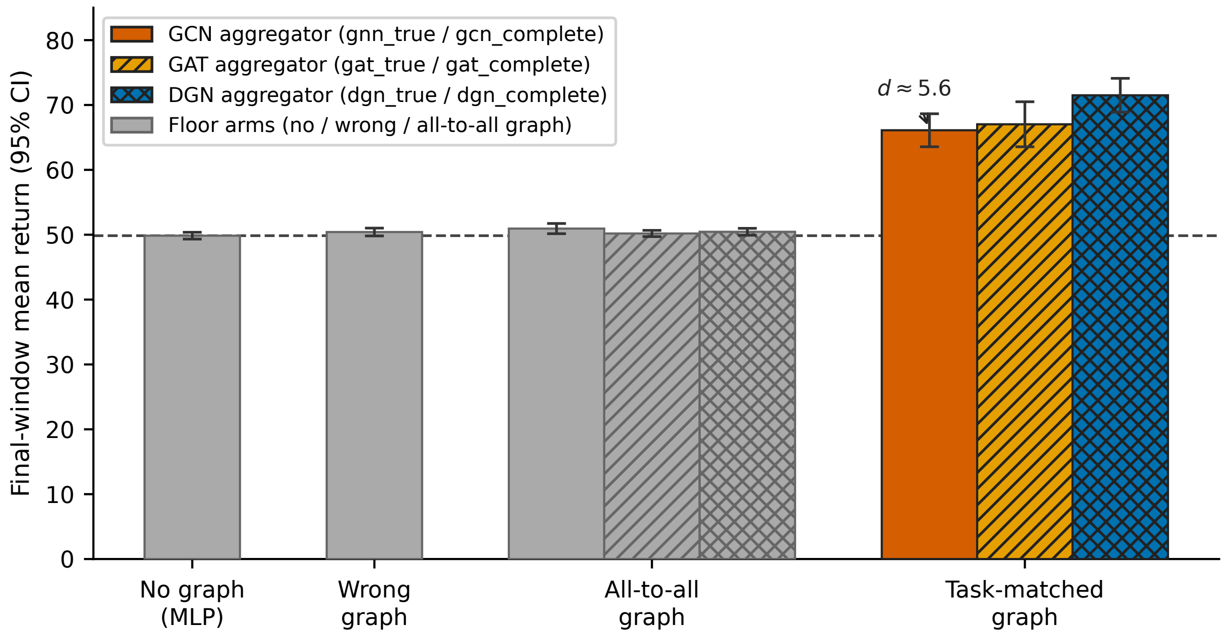


Figure 11: **It is the structure, not the channel.** Final-window mean return (95% CI, 12 seeds) on the pair-routing task (N=6, ego observations, 30k steps). Only the *task-matched* graph rises well above the random floor (∼50): the no-graph control (`mlp`) and a density-matched wrong graph sit at the floor, the all-to-all graph (the *most* communication) one point above it, while the task-matched graph reaches 66–71 depending on the aggregator (d≈5 vs. the floor). Learned attention (GAT, DGN) helps only *on* the matched graph; over all-to-all it floors like the fixed GCN. The right edges, not more edges, carry the advantage.

	N=4	N=6	N=8	N=10
`gnn_true`−`mlp` (return)	+11.8	+16.3	+16.4	+11.0
Cohen's d	3.3	5.6	3.0	1.6
fractional lift	35.5%	32.6%	24.7%	13.1%

Effect size and fractional lift decline past N=6. We therefore report the "grows-with-N" hypothesis as **not supported**, and treat N as a robustness axis (the boundary holds across team sizes) rather than as a dose-response. *A caveat on the cross-N rows:* total reward, and hence the floor, grows with team size (the `mlp` floor is 33.3, 49.9, 66.6, 83.7 at N=4, 6, 8, 10 as more agents earn incidental co-location reward), so neither the absolute return gap nor the "fractional lift" denominator is directly comparable across columns. We lean on Cohen's d (scale-invariant) as the primary attenuation evidence; the non-monotone conclusion holds under either.

7.4 Robustness to observability

The partner's goal is private under *any* observation mode, so the win should not be a partial-observability artefact—and it is not. Under *full* observations (N=6, 12 seeds) the effect is *larger*: `gnn_true` beats the no-graph control by +22.6 (d = 5.4), with the `gnn_true` vs. `gnn_wrong` and `gnn_true` vs. `gnn_complete` contrasts at the same magnitude. Adding observability does not let the rival arms route the private information; only the matched edges do. One input caveat is disclosed: under the full-observation mode the neighbour-displacement block of each agent's observation follows its arm's communication graph (`coord_grid.py`), so—unlike the ego-mode cells, where all arms' inputs are byte-identical—the graph arms here differ in inputs as well as edges. The input-matched contrast in this cell is `gnn_true` vs. `mlp` (both observe the true-matching neighbour block), which is the $++22.6$ comparison we quote; we rest the full-obs robustness claim on it.

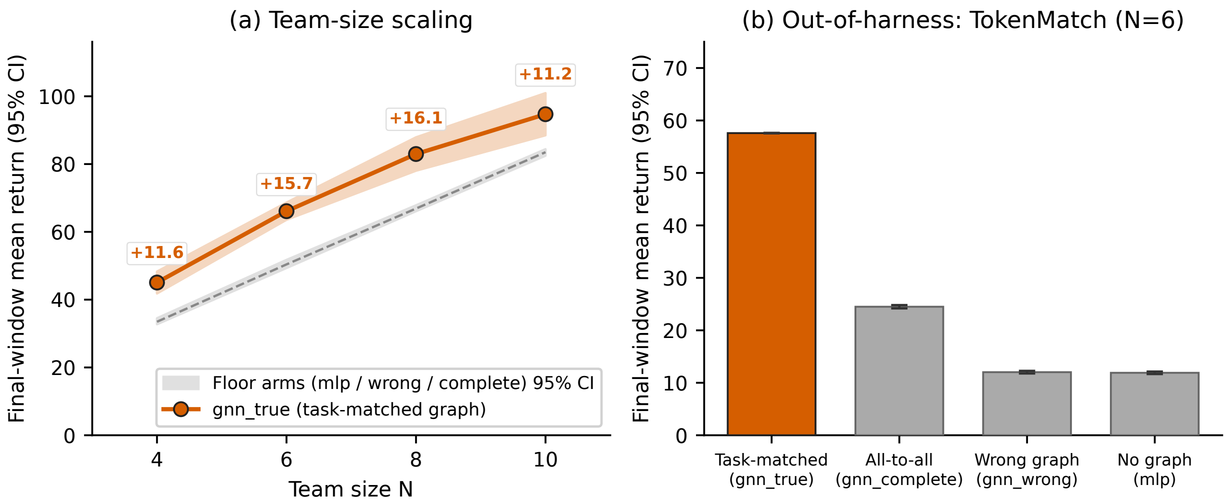


Figure 12: **Left: the advantage holds across team size but is non-monotone.** Task-matched `gnn_true` (solid line) stays above the control floor (shaded band: `mlp`/wrong/all-to-all 95% CI) at every $N \in \{4, 6, 8, 10\}$; the gap over the no-graph floor (labelled) peaks near N=8 rather than growing monotonically. **Right: out-of-harness replication** on TokenMatch (shown standalone, with discussion, as Figure 14 in §7.7). All bars/points are 95% CIs over 12 seeds.

7.5 Genuine topology: the effect survives at degree-2, much smaller

A matching is degree-1 (point-to-point), which tests *routing* more than *topology*. To test topology we use `nbr_routing` (degree-2 ring): each agent must reach the *centroid of its two ring neighbours'* goals, with `gnn_true` = ring, `gnn_wrong` = skip-ring (± 2, matched degree), and `gnn_complete` = all-to-all. Under the matched protocol (N=6, 12 seeds, 30k), `gnn_true` (52.67) significantly beats the wrong topology (+2.6, $d = 1.7$), all-to-all (+2.2, $d = 1.4$), and no communication (+2.4, $d = 1.5$), all at $p_{\text{Holm}} < 0.005$ and Mann–Whitney $p < 0.005$. So the structure/topology effect *survives* when the receptive field is a neighbourhood set rather than a single partner—but at the pre-registered 30k budget it is about 7× smaller than at degree-1 (+2.4 / ∼5% vs. +16 / ∼33%; Figure 13). We treat 7× as an *upper bound* on the attenuation: as the footnote records, exploratory longer-budget runs widen the degree-2 gap, so the genuine-topology effect may be materially less attenuated than the 30k cell shows. A fixed mean-aggregating GCN exploits a neighbourhood-set (centroid) task far less efficiently than single-partner routing.[2]

7.6 Learned attention does not substitute for structure either

A natural question is whether an expressive *learned* aggregator over the *complete* graph could rediscover the right edges and thus substitute for fixed structure. We add single-head (GAT) and multi-head (DGN/G2ANet-style) attention arms, **stabilisation-matched** to the GCN (the same residual+LayerNorm), N=6, 12 seeds—so the earlier confound, attention arms lacking the GCN's stabilisation, is removed. All quoted p-values are Holm-corrected over the six-member `gnn_true`-vs-each-rival family (the artefact `struct_verdict_pair_N6_ego_attention.csv`; we label raw vs. Holm-corrected explicitly). Because this attention sweep re-corrects over its own six-member family, the `gnn_true` vs. `gnn_complete` contrast's p_{Holm} here (6.6×10^{-8}) differs slightly from the core three-contrast value in Table 2 (3.3×10^{-8}); both are the correct per-family Holm value and the conclusion is identical. Two facts settle the question. *(i) The attention arms are not under-trained on the true graph:* given the *right* graph, learned attention is at least as good as the GCN—single-head GAT is statistically indistinguishable (`gat_true` 67.03 vs. `gnn_true` at 66.10; gap +0.9, n.s., raw Welch $p = 0.64$; 95% CI on the gap $[-3.2, +5.0]$, so equivalence is bounded rather than proven),

[2]Exploratory (non-confirmatory) 3-seed runs at 80k steps show the same arm ordering with a larger gap ($\sim+5$), suggesting the 30k confirmatory cell is not at plateau; only the pre-registered 12-seed / 30k cell above is treated as the measurement.

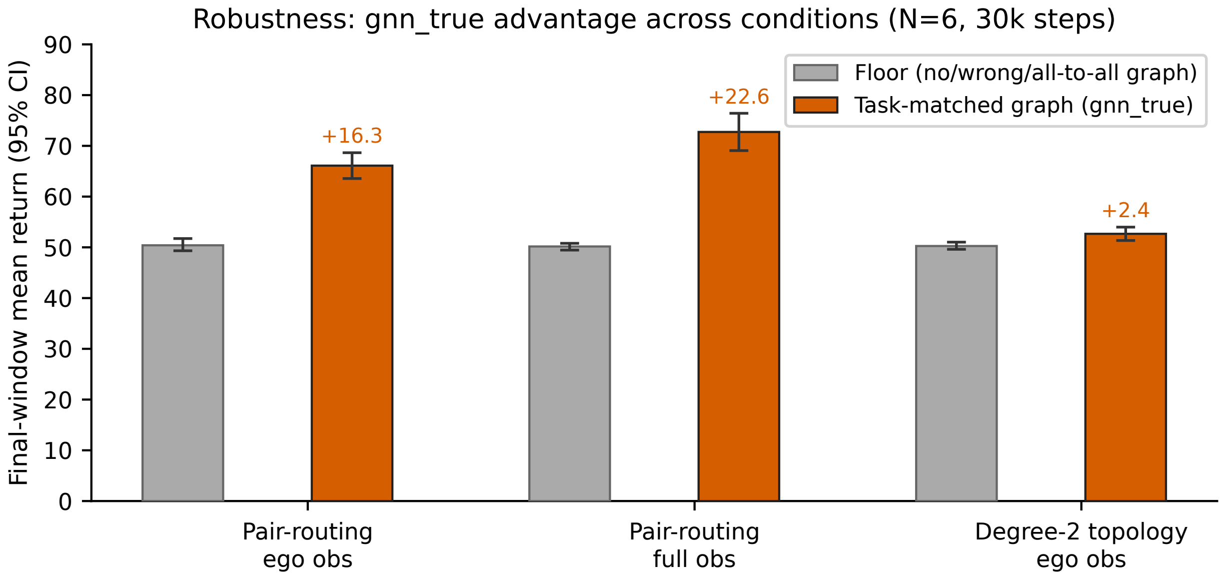


Figure 13: **The advantage is robust across conditions.** Task-matched `gnn_true` (orange) versus the control floor (gray: no/wrong/all-to-all graph), 95% CI over 12 seeds at N=6 (30k steps). The structure advantage holds under ego observations (+16.3) and is *larger* under full observability (+22.6); at genuine degree-2 topology it survives but shrinks sharply (+2.4), consistent with a fixed mean-aggregator exploiting a neighbourhood-set (centroid) task far less efficiently than single-partner routing.

and multi-head DGN actually *exceeds* it (`dgn_true` 71.50, +5.4 over `gnn_true`, Welch $p_{\text{Holm}} = 0.007$; we flag this last comparison as *exploratory*—the pre-registered H-attn asked only whether attention-over-complete *reaches* `gnn_true`, not whether attention-over-true exceeds it, so the +5.4 direction is reported descriptively, not as a confirmatory finding). *(ii) Over all-to-all, both attention mechanisms still floor within budget:* `gnn_true` beats `gat_complete` (50.19) and `dgn_complete` (50.45) by +15.9 and +15.7 respectively (Cohen's $d{\approx}5.5$, $p_{\text{Holm}} < 10^{-4}$). One asymmetry qualifies this clause: the complete-graph attention arms face a *strictly harder* optimisation than their true-graph siblings—under the true graph the partner is the only neighbour and attention has nothing to discover, whereas under all-to-all the head must learn to attend to one partner out of $N-1$ from agent-id one-hots. Because that harder optimisation could in principle just be *slower*, we pre-registered and ran the decisive robustness check: the identical attention sweep at the 5× (150k-step) budget *matched to the negative endpoint*, so no budget asymmetry favours either side (`results/phaseC_150k/`, 12 seeds; decision rule frozen before results). The conclusion does not merely survive—it strengthens. At 150k the complete-graph attention arms rise only *marginally* above the no-communication floor—`gat_complete` 51.08 and `dgn_complete` 53.26 vs. the 49.96 floor, i.e. +1.1/+3.3, under a tenth of the 45-point gain the task-matched arms make—while `gnn_true` itself keeps learning (66.10 → 94.96), so the gap *grows* from $d{\approx}5.5$ at 30k to $d{\approx}12$ at 150k (`gnn_true` over `gat_complete`/`dgn_complete` by +43.9/+41.7, $p_{\text{Holm}} < 10^{-4}$, complete seed separation). That the complete arms move *at all* (`dgn_complete` climbs 50.1 → 53.3 across quintiles) is itself informative: gradients do flow through the all-to-all path, so this is a *live* control that plateaus far below—not a silently broken one—which rules out the "the complete-graph arm never optimised" objection. The *same* attention architectures train well at this budget *given the right graph*—`gat_true` 96.71, `dgn_true` 112.49, both far above the floor—which is the validity gate that licenses the reading: the complete-arm floor is structural, not a lack of budget. The learning curves make this direct (Table 3): across all five training quintiles the complete arms stay within a few points of the floor while every true-graph arm climbs steeply, so "attention over all-to-all merely needs more training" is refuted by measurement, not deferred. We bound the claim to the budgets we tested (30k and 150k) rather than asserting impossibility in principle. The result stands: an expressive, well-trained attention aggregator does *not* recover the routing advantage from the wrong structure within the same budget that suffices given the right graph—so expressiveness *ampli-*

Table 3: **Attention over all-to-all stays near the floor—at the 5× budget matched to the negative endpoint.** Mean return by training-progress quintile (Q1–Q5), averaged over 12 seeds, on PairRouting (N=6, ego), 150k steps (`results/phaseC_150k/`; pre-registered F1 check). Every *true-graph* aggregator—GCN, GAT, and multi-head DGN—climbs steeply; the *same* attention architectures over the *complete* graph rise only marginally above the no-communication floor (49.96) over the entire budget (`gat_complete` +1.1, `dgn_complete` +3.3), remaining 40+ points below the task-matched arms. The all-to-all deficit is therefore structural, not under-training.

Arm	Graph	Q1	Q2	Q3	Q4	Q5
`gnn_true`	matching	50.9	57.5	68.5	83.4	**95.0**
`gat_true`	matching	51.0	57.9	68.9	83.3	**96.7**
`dgn_true`	matching	51.2	64.9	82.7	99.7	**112.5**
`gat_complete`	all-to-all	50.0	50.1	50.1	50.4	51.1
`dgn_complete`	all-to-all	50.1	50.1	50.3	51.3	53.3
`mlp` (no comm.)	—	50.0	50.1	50.1	49.9	50.0

Table 4: **Out-of-harness replication** (TokenMatch, a non-spatial referential matching game; N=6, K=5 tokens, 12 seeds). Same structure-isolating arms as PairRouting (only `adjacency()` differs). The task-matched graph near-solves the task; the wrong graph and no-communication sit at the random floor (≈12); all-to-all routes only partially because the partner's token is one of the averaged inputs. We report this as complete distributional separation (the `gnn_true` seeds do not overlap any rival); under the near-zero within-arm variance of this near-deterministic constructed task, a Cohen's d is astronomically large and *uninformative* as an effect magnitude, so we do not headline it.

Arm	comm. graph	final return [95% CI]
`mlp`	none	11.90 $[11.67, 12.14]$
`gnn_wrong`	wrong matching	12.03 $[11.78, 12.27]$
`gnn_complete`	all-to-all (partial routing)	24.50 $[24.18, 24.81]$
`gnn_true`	**the task matching**	**57.59** $[57.55, 57.63]$

fies structure (DGN exceeds the GCN by +5.4 given the right edges) but does not *substitute* for it. The advantage is the structure, not the channel or the capacity, and is not recovered by expressiveness alone.[3]

7.7 Out-of-harness replication: not a gridworld artefact

Everything above is CoordGrid, a navigation task; the effect could be a gridworld artefact. To rule that out we replicate the structure test in a *completely different* environment—TokenMatch, a referential matching game with *no grid, no movement, no navigation*: each agent privately holds a token and must *output its partner's token*, which can reach it only over the graph. The structure-isolating arms carry over unchanged (only `env.adjacency()` differs): `gnn_true` (the matching), `gnn_wrong` (a wrong matching), `gnn_complete` (all-to-all), and the no-communication `mlp` control. With N=6, K=5 tokens, and 12 seeds (Table 4; Figure 14), `gnn_true` *near-solves* the task (57.59 of a maximum 60, ≈96%) and is completely separated from every rival—it beats the wrong matching by +45.6, all-to-all by +33.1, and no communication by +45.7 ($p < 10^{-4}$ on both tests, variance near zero). All-to-all does *partial* routing here (24.50, ≈+12 over the ≈12 floor)—the partner's token is one of the inputs the mean aggregator averages—but it remains +33.1 below the task-matched graph. The structure effect therefore reproduces, more cleanly, in a structurally unrelated task: it is **not** a CoordGrid artefact.

[3]The degree-2 (`nbr_routing`) attention cell tells the same story at 12 seeds: `gnn_true` 52.7, `gat_true` 52.3, and `dgn_true` 53.4 all sit together above the floor, while `gat_complete`/`dgn_complete` floor at ∼50.4 alongside the no-communication control (50.2)—so on the degree-2 topology too, attention helps only on the right graph and floors over all-to-all. (An earlier 3-seed run of this cell was underpowered; the 12-seed cell reported here is the confirmatory measurement.)

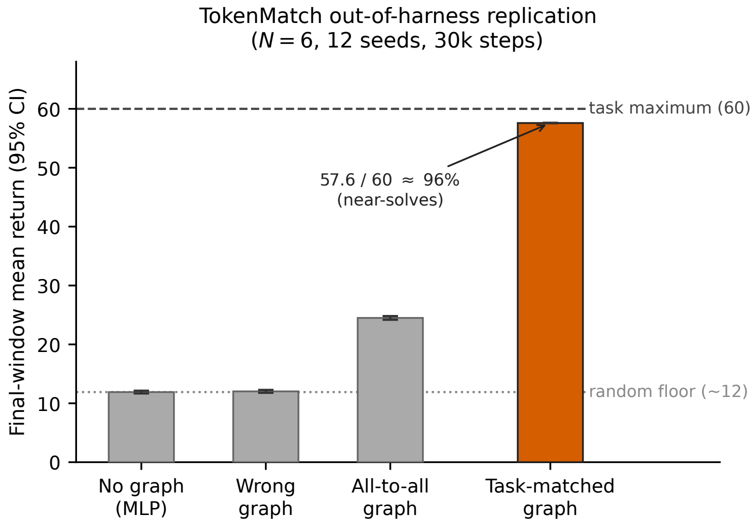


Figure 14: **Out-of-harness replication.** Final-window mean return (95% CI, 12 seeds, 30k steps) on TokenMatch (N=6, K=5 tokens), a non-spatial referential game with no grid, movement, or navigation. The task-matched graph near-solves the task (57.6 of a 60 maximum, ≈96%), completely separating from the wrong-matching graph (12.0) and the no-graph control (11.9), both at the random floor. All-to-all achieves partial routing (24.5; the partner's token is one of the averaged inputs) but remains far below the task-matched graph. Confirms the structure effect of Figure 11 is not a gridworld artefact.

Scope of the positive endpoint. These tasks are *built* so that routing is necessary—the right design to isolate structure from capacity, convention, and observability, and the pre-registered plan. The result *coexists with* the negative endpoint; it does not show the negative was wrong. The positive effect now reproduces out of harness in a structurally unrelated, non-spatial environment (TokenMatch, §7.7), so it is not a gridworld artefact; both TokenMatch and PairRouting are nonetheless purpose-built cooperative tasks with a private-partner-routing structure, and replication on a *standard* third-party benchmark (an MPE/LBF variant with private per-agent goals) remains future work—we do not generalise beyond "task-structured routing under partial information." For reproducibility, the N=6 headline is *bit-identical* across two independent runs spanning a script refactor—genuine seed determinism, not relabelled copies.

8 Discussion

When does graph structure help? The boundary. The answer is neither "always" nor "never" but a boundary with two identified endpoints. *At the negative endpoint*—full-information or convention-reducible tasks, Sections 6.2–6.5—the GNN-QMIX encoder does *not* outperform the structurally identical QMIX. The capacity control (Section 6.5) shows this is not because the extra parameters are wasted: MLP-QMIX, carrying the same parameters without the graph, matches QMIX, isolating the effect on the graph aggregation, which confers no benefit and—in its common *unstabilised* form—is actively harmful on CoordGrid and MPE, survives graph attention, and grows out to N=16. That active harm is an optimisation pathology of the unstabilised operator, not an intrinsic cost: a stabilised GCN recovers to parity with the controls (Appendix H), so the claim that survives for the stabilised operator is the weaker "no benefit" rather than "harm." On standard benchmarks usually credited to graph methods the picture is the same *no-benefit-to-harm* one: on Level-Based Foraging the effect tracks team size—neutral at two agents but significantly below QMIX at three ($d = -2.24$, Welch $p = 0.014$; Appendix F). We note that this significant LBF result is the (confounded) GNN-QMIX-vs-QMIX comparison; the parameter-matched GNN-QMIX-vs-MLP-QMIX contrast that our method relies on points the same way ($d = -1.11$) but is not powered for significance at n=5, so on LBF we treat the identified penalty as directional, not confirmed. (A single-seed StarCraft II `3m` check is reported in Appendix F as a non-inferential anecdote, not as evidence—

one seed admits no variance estimate.) In this full-information / convention-reducible regime the measured effect ranges from neutral to harmful and is never positive. *At the positive endpoint* (Section 7), where the task forces an agent to act on private information held by one specific partner it cannot observe, the stabilised graph operator helps—and decisively so ($d \approx 5$)—*but only when its topology matches the task's coordination edges*. This is the *same* stabilised operator that Appendix H shows is neutral (parity with the no-graph control) at the negative endpoint: so the boundary is not an artefact of comparing two different operators. The same stabilised GCN is inert under full information and a large asset under task-structured routing—the discriminator is the task's information structure, not the architecture. The two readings coexist: a graph earns its keep exactly when it is the only path for task-structured information *and* points at the right neighbours, and is neutral (or, unstabilised, harmful) otherwise. *This is a claim about a mechanism, not a verdict on any system*: we test the relational operators these methods rely on (a fixed GCN and learned graph attention) inside one controlled QMIX-style harness on small-to-moderate cooperative tasks, not the full published systems at SMAC scale. Within that scope it is a far more constrained reading of graph-based value factorisation than a *regime-agnostic* reading of these results (Jiang et al., 2020; Liu et al., 2020; Naderializadeh et al., 2020) would suggest. As stressed in the introduction, DGN and G2ANet report their gains in *partially-observable* settings—precisely the regime our *positive* endpoint endorses—so our constraint falls on the *full-information* generalisation of those results, not on the regimes those papers actually evaluate; this is consistent with the evaluation-rigour concerns of Henderson et al. (2018) and Agarwal et al. (2021).

The practical rule. Add a coordination graph, and route along *task-relevant* edges rather than all-to-all, only when agents must act on information held by specific others they cannot observe. The controls make the rule sharp, and we state it in the form the data support rather than as an absolute: with a fixed-budget learned aggregator, all-to-all recovers *at most partial* routing and never approaches the task-matched graph (≈7% of the gap on PairRouting—about one point above the no-communication floor, a statistically detectable sliver of routing, §7.2—and ∼27% on TokenMatch, where the partner's token is one of the averaged inputs)—and a density-matched wrong graph never beats no graph at all. The difference between the two tasks is exactly that in TokenMatch the partner's signal survives mean-pooling enough to be partly usable, yet even there broadcasting loses by −33. Pointing the channel at the wrong neighbours wastes it; broadcasting to everyone does not rescue it. Where the task is full-information or convention-reducible, prefer a state-conditioned mixer (QMIX) and add no graph—the relational bias there is at best inert, and if implemented as the standard unstabilised GCN, a liability.

An important caveat on actionability. This rule presupposes the practitioner *knows* (or can reliably estimate) the task's true coordination graph; learning the graph from scratch is out of scope here. A complementary line of work attacks exactly this gap by *learning* the coordination graph rather than assuming it—e.g. DICG (Li et al., 2021) and CASEC (Wang et al., 2022)—and our controls quantify the cost of the mis-specification those methods aim to avoid; characterising whether such learned-graph methods recover the task-matched structure under our wrong-graph/all-to-all isolation is natural future work. That is not a minor proviso, because our all-to-all result shows the obvious fallback fails: a learned dense channel does *not* trivially recover the benefit (all-to-all, and even a learned attention head over all-to-all, stay near the floor even at 5× the training budget—§7.2, §7.6), and a wrong graph is no better than no graph. Graph *mis-specification* therefore carries a real cost, and the safe default where the coordination structure is unknown or uncertain is the no-graph state-conditioned mixer: in that regime the negative endpoint dominates. A further practical bound: even a near-perfectly specified graph leaves most of the task on the table—`gnn_true` captures only ∼16% of the routing headroom on PairRouting versus ∼96% on TokenMatch (§7.2, §7.7), because PairRouting layers gridworld navigation and credit assignment on top of routing while TokenMatch removes navigation; the gap a graph closes is the *routing* bottleneck, not the whole task. Finally, expressiveness does not let you skip specifying the graph: the most expressive aggregator we test (multi-head attention) trains *better* given the right edges (`dgn_true` 71.50, an exploratory +5.4 over the GCN, $p_{\text{Holm}} = 0.007$) yet still floors over all-to-all (`dgn_complete` 50.45)—the cleanest single rebuttal to "the GCN just under-fits" and "mean-pooling dilutes the signal," since the strongest aggregator both fits better and still needs the right graph.

The depth–diameter heuristic does not survive the data. The pre-registered H3 prediction (choose GNN depth L near the graph diameter d) is falsified at the family level (1 of 3 diameters passes; Section 6.4) and replaced by a sharper pattern: $L \in \{1, 2\}$ beats $L \in \{3, 4\}$ by a wide margin across every diameter, consistent with over-smoothing (Li et al., 2018) and over-squashing (Alon & Yahav, 2021) but more aggressive here than the receptive-field intuition predicts. The rule: in small-team CTDE with modest replay budgets, do not stack more than two GCN layers before a QMIX-style mixer.

When *not* to use a GNN mixer. Three cases. (i) The task has no real coordination graph, so the bias buys nothing and pays the cost of representing (or mis-specifying) one. (ii) The diameter d exceeds the feasible depth L, so the GNN cannot route information across the team and a state-conditioned mixer (QMIX) fits better. (iii) The case our experiments document mechanistically: when the task is full-information or reducible to a global convention, so no agent depends on another's private information, the graph mixer buys nothing over a parameter-matched no-graph control—and if implemented as the standard unstabilised GCN it is *worse*, both less accurate and far less stable across seeds than QMIX (the capacity control pins this on the graph aggregation, not the parameters; stabilisation removes the instability but not the futility, Appendix H). (The discriminator is the information structure, not team size or diameter: our positive endpoint is equally small-team and low-diameter, yet there the graph helps.) Practitioners should not assume "a coordination graph exists" implies "a graph-conditioned mixer will help."

When *to* use one. The positive endpoint marks out the complementary case: a graph mixer pays off when (i) the task requires an agent to act on information held by a specific other it cannot observe, so the graph is the only route for that information, *and* (ii) the communication topology is aligned with the task's coordination structure. Our controls show both conditions are necessary, not just sufficient: in the gridworld headline (Section 7.2), removing alignment (the wrong graph) or flooding the channel (all-to-all) each collapsed the advantage to near zero, and in the token-matching task all-to-all retained only weak, task-incidental routing—far below the matched graph in both. The same stabilised aggregation that is merely *inert* in the convention-reducible regime (parity with the control, Appendix H; and a liability only in its unstabilised form) becomes a large, reproducible asset here ($d \approx 5$), so the design question is not "graph or no graph" but "do the task's information dependencies match the edges I wire in."

Implications for benchmark design. Because which algorithm wins is a function of the graph regime rather than of the algorithm, single environment / single-graph comparisons are condition-dependent; future cooperative-MARL benchmarks should expose the coordination graph as a controllable axis, in the spirit of CoordGrid.

9 Limitations

We list explicit limitations rather than burying them.

1. **Scale.** Our largest team is N=16 on a tabular-flavoured gridworld; $N \gg 16$ and visually complex tasks (SMAC-scale) are out of scope. The penalty *grows* out to N=16 (Appendix J), so scale within our range strengthens rather than threatens the result. (Step budget is also moot: decisive cells run at 150,000 steps, 5× the main sweep.)
2. **Coordination-graph design.** CoordGrid ties reward to graph edges directly, the cleanest but most graph-favourable test. The MPE `simple_spread` replication (Section 6.3), where the graph is an unprivileged k-NN proximity graph, mitigates this; but tasks whose true coordination structure matches no observable graph may differ.
3. **GNN family and operator variants.** We test a fixed GCN (plain and stabilised), single-head (GAT-QMIX) and multi-head dot-product (DGN-QMIX) attention (Appendices H, I); at the negative endpoint no variant recovers a benefit over the no-graph control. We isolate one operator effect (residual+LayerNorm converts the plain GCN's active harm to mere parity) but do not sweep the full stabilisation/normalisation design space, and learned message passing (GIN) and edge-feature GNNs remain open.
4. **Positive endpoint is still a synthetic-task result.** The positive boundary (Section 7) is a constructed-task existence/boundary result. It now replicates out of harness in a structurally unrelated, non-spatial

environment (TokenMatch, Section 7.7), so it is not a CoordGrid artefact; but both TokenMatch and PairRouting are purpose-built cooperative tasks with a private-partner-routing structure. Replication on a standard third-party benchmark (an MPE/LBF variant with private per-agent goals) remains future work, and we do not generalise beyond "task-structured routing under partial information."

5. **Mis-specification is sampled only at the endpoints.** We contrast an exactly-correct graph against a fully-wrong matching and all-to-all, but not the intermediate case a practitioner faces—a graph with a *fraction* of its partner edges correct. We therefore cannot say whether the benefit degrades gracefully or cliffs as the graph drifts from correct; a partial-correctness sweep (edges correct in $\{0, 0.25, 0.5, 0.75, 1\}$) would turn the binary right-vs-wrong contrast into a graded mis-specification-cost curve and is the cheapest decision-relevant extension.

6. **Single-CPU training.** We did not retune learning-rate / batch-size choices for GPU throughput.

7. **Seed count.** The main CoordGrid and MPE sweep runs at 10 seeds, at or above the ≥ 5 recommended (Henderson et al., 2018; Agarwal et al., 2021); the LBF check uses five seeds and the preliminary SMAC check a single seed (Appendix F).

8. **Fixed-horizon bootstrap.** Episodes are truncated with `done=True`, biasing the boundary value toward zero, the standard QMIX choice, applied identically to every algorithm, so cross-algorithm comparisons are unaffected.

10 Conclusion

We presented a controlled study of graph-based value factorisation in cooperative MARL that locates *both* endpoints of a boundary, holding the per-agent encoder, mixer, optimiser, and exploration schedule fixed and adding a parameter-matched no-graph control (MLP-QMIX). *Negative endpoint:* in full-information / convention-reducible tasks GNN-QMIX does not beat QMIX, and the control *identifies* the source as the graph aggregation rather than the added capacity. In its common unstabilised form the graph is actively harmful ($d = -3.9$ at N=8; significant under Mann–Whitney and Holm-corrected Welch)—the deficit survives 5× training, grows monotonically to N=16, and inflates across-seed variance by roughly an order of magnitude—but this harm is an optimisation pathology: stabilising the GCN (residual+LayerNorm) recovers it to parity with the control (Appendix H), so the robust claim is *no benefit*, with active harm scoped to the unstabilised operator. The graph is never significantly positive on MPE `simple_spread` (penalty visible under the capacity decomposition; naive comparison null) or on Level-Based Foraging (significantly below QMIX at three agents; matched-control contrast directional at n=5), and does not recover a benefit under graph attention; the pre-registered $L \approx d$ heuristic is falsified in favour of the blunt rule "keep GCN depth ≤ 2." *Positive endpoint:* on a constructed pair-routing task where each agent must reach a partner's private goal, the task-matched graph beats all-to-all communication—the only rival that carries the partner's information, yet it recovers almost none of it—at byte-identical parameters and observations ($d \approx 5$ at degree-1; Welch/Holm and Mann–Whitney $p < 10^{-4}$, complete seed-level separation at n=12), while a density-matched wrong graph (which carries no partner information by construction) floors with the no-graph control—more communication did not help, the *right* communication did. The advantage is structure, not the channel or the capacity, and is not recovered by expressiveness alone: neither the fixed mean-aggregator nor a stabilisation-matched learned attention head recovers the signal from all-to-all, even at 5× the training budget, while given the right graph a more expressive aggregator helps (+5.4), so expressiveness amplifies structure but cannot substitute for it. The effect is significant at every N=4–10 (non-monotone), robust to observability, survives at degree-2 about 7× smaller at the 30k budget (an upper bound on the attenuation), and replicates out of harness in a non-spatial token-matching game (TokenMatch), so it is not a gridworld artefact. The practical rule that unifies the two: route along task-relevant edges, not all-to-all, only when agents must act on information held by specific others they cannot observe. The positive endpoint remains a synthetic-task result—replication on a standard third-party benchmark with private per-agent goals is the natural next step. We release the harness and the MLP-QMIX control to make such measurements cheaper.

Broader Impact Statement

This work is a controlled empirical study of when coordination-graph structure helps cooperative multi-agent reinforcement learning, conducted entirely in synthetic environments (gridworlds, a referential game, and standard research benchmarks). Its contribution is measurement and methodology—matched controls and pre-registered comparisons for evaluating graph-based coordination—rather than a deployable capability, and we foresee no direct societal risk beyond those generic to improved multi-agent coordination. If anything, the negative endpoint cautions practitioners against adding relational machinery on the assumption that it helps, which may reduce wasted computation.

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A Training loop

B Hyperparameters and compute

Compute envelope. All experiments ran on a single CPU (commodity hardware, PyTorch CPU build). No GPU was used. CoordGrid is light enough that this is not a bottleneck for the team sizes we report; it does cap the feasible N at the upper end (see Section 9). Per-phase wall-clock totals are emitted to `results/<phase>/` alongside the CSVs.

Execution environment. Experiments and analysis ran CPU-only on commodity x86-64 Windows hosts (primary machine: 24-thread Intel Raptor-Lake desktop CPU, 64 GiB RAM, Windows 11) under Python 3.13 with the CPU build of PyTorch 2.12, NumPy 2.5, SciPy 1.18, and pandas 3.0; the per-run `config.json` files in the artifact record the exact configuration of every run. Statistical note: Mann–Whitney U p-values are SciPy's asymptotic (normal-approximation) values; the exact two-sided floor at 12-vs-12 seeds is $2/\binom{24}{12} \approx 7.4 \times 10^{-7}$, so quoted MWU values at or below 10^{-4} indicate complete rank separation rather than a precise tail probability.

Algorithm 1 Shared training loop (CTDE, off-policy, Double-DQN). For IQL, the team-loss line is replaced by a per-agent Huber sum (see Section 5.2).

Require: environment $\textsc{Env}$, algorithm $\textsc{Alg}$ (online + target nets, mixer), total steps T, batch size B, target-update interval τ, exploration schedule $\epsilon(t)$, update period k, warmup t_{w}. ($\textsc{Env}$ and $\textsc{Alg}$ are set apart in small caps so that $\mathcal{A}$ and $\mathcal{E}$ keep their body-text meanings: action set and edge set.)

Initialise replay buffer $\mathcal{D}$, target net $\theta^- \leftarrow \theta$.
$(o, s) \leftarrow \textsc{Env}.\textsc{reset}()$; $A \leftarrow \textsc{Env}.\textsc{adjacency}()$
for $t = 1, \ldots, T$ **do**
    $a \leftarrow \textsc{Alg}.\textsc{act}(o, A, \epsilon(t))$ ▷ ϵ-greedy per-agent, params shared
    $(o', s', r, \mathrm{done}) \leftarrow \textsc{Env}.\textsc{step}(a)$
    $\mathcal{D}.\textsc{add}((o, s, a, r, o', s', \mathrm{done}, A))$
    if $t \geq t_{\mathrm{w}}$ **and** $t \bmod k = 0$ **then**
        batch $\sim \mathcal{D}$ (size B)
        $\mathbf{Q} \leftarrow \textsc{Alg}.Q(o, A; \theta) \in \mathbb{R}^{B \times N \times |\mathcal{A}|}$ ▷ per-agent Q over all actions
        $q_a \leftarrow \mathrm{gather}(\mathbf{Q}, a) \in \mathbb{R}^{B \times N}$; $Q_{\mathrm{tot}} \leftarrow f_{\mathrm{mix}}(q_a; s)$
        $a^* \leftarrow \arg\max_{a'} \textsc{Alg}.Q(o', A; \theta)$ ▷ Double-DQN action selection, online net
        $\bar{q} \leftarrow \mathrm{gather}(\textsc{Alg}.Q(o', A; \theta^-), a^*)$; $Q^-_{\mathrm{tot}} \leftarrow f^-_{\mathrm{mix}}(\bar{q}; s')$
        $y \leftarrow r + \gamma(1 - \mathrm{done}) \cdot Q^-_{\mathrm{tot}}$
        $\theta \leftarrow \theta - \eta \nabla_\theta \mathrm{Huber}(Q_{\mathrm{tot}}, y)$ ▷ Adam, grad-norm clip 10
        if $t \bmod \tau = 0$ **then** $\theta^- \leftarrow \theta$
        end if
    end if
    if done **then** $(o, s) \leftarrow \textsc{Env}.\textsc{reset}()$; $A \leftarrow \textsc{Env}.\textsc{adjacency}()$
    else $(o, s) \leftarrow (o', s')$
    end if
end for

Table 5: Shared training hyperparameters across all algorithms and all phases. Values were chosen once at Phase 0 and held fixed; no algorithm-specific tuning was performed. Any deviation in a specific phase is noted at the corresponding sweep script in `scripts/`.

Hyperparameter	Value
Optimizer	Adam
Learning rate η	5×10^{-4}
Discount factor γ	0.99
Loss	Smooth-L1 (Huber, $\beta = 1.0$)
Per-agent encoder hidden	64×64 ReLU MLP
QMIX mixer embedding	32
QMIX hypernet hidden	32
GCN hidden dimension	64
GCN normalization	Symmetric $\hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$ (per batch element)
Replay buffer capacity	5×10^4 transitions
Batch size	32
Warmup before first update	200 env steps
Update frequency	every 4 env steps
Target hard-update interval	every 200 gradient steps
Gradient norm clip	10.0
Exploration (ϵ)	linear 1.0 $\rightarrow$ 0.05 over 80% of run
Double-DQN target	yes
Parameter sharing	yes (agent-id one-hot appended to obs)
Random seeds per condition	10 (Phases 1–4, 8–11); 12 (Stage C / positive endpoint)

C Reproducibility

The complete source code (environments, algorithms, training loop, tests), the per-run configuration of every run (seed, hyperparameters, budget), all aggregate and per-contrast statistics CSVs, and the exact commands used to produce every table and figure in this paper (`docs/REPRO.md`) are provided as anonymized supplementary material. Each run is fully seeded and deterministic on CPU; the CSVs under `results/` are the direct inputs to the analysis and figure scripts under `scripts/`. Where a lower-seed smoke directory coexists with the confirmatory data, the canonical directories are the ones the paper reads: `results/phase1_s10`, `results/phase2_s10`, and `results/phase4_s10` (10 seeds) rather than the 3-seed `results/phase1`, `phase2`, `phase4` smoke sets, and `results/phaseC/phaseD` for the Stage C/D cells; `docs/REPRO.md` lists the mapping per table and figure.

D Compute scaling decisions

Table 6: Compute budget per experiment: pre-registered vs. actually executed (steps × seeds × algorithms/arms × conditions). Wall-clock totals, where measured, are on a single commodity CPU (PyTorch CPU build); no GPU was used. The Phase 3 MPE row is the one large executed-below-registration deviation; its consequences are discussed in Section 6.3.

Experiment	Pre-registered	Executed	Reason for deviation
1 (pilot)	$5 \cdot 10^4 \times 3 \times 4$	$5 \cdot 10^4 \times 10 \times 4$	seeds raised 3→10.
2 (E1 sweep)	$2 \cdot 10^5 \times 5 \times 4 \times 3 \times 2$	$3 \cdot 10^4 \times 10 \times 4 \times 2 \times 2$	CPU envelope; seeds 5→10.
3 (MPE)	$\geq 10^6 \times 5 \times 4 \times 2$	$1.5{\cdot}10^5 \times 10 \times 5 \times 2$	CPU envelope (15% of registered steps).
4 (ablation)	$\geq 5 \cdot 10^4 \times \cdots$	$2 \cdot 10^4 \times 10 \times 4 \times 3$	CPU envelope; seeds → 10.
8 (capacity control, ring)	$\geq 10^5 \times 5$	$1.5 \cdot 10^5 \times 10 \times 5 \times 2$	seeds and steps *raised*.
8 (line topology)	— (post-hoc robustness)	$1.5 \cdot 10^5 \times 10 \times 5 \times 2$	as ring.
10 (attention, CoordGrid+MPE)	— (post-hoc robustness)	$1.5 \cdot 10^5 \times 10 \times 2 \times 4$	as Phase 8.
11 (scaling $N \in \{12, 16\}$)	— (post-hoc robustness)	$1.5 \cdot 10^5 \times 10 \times 3 \times 2$	as Phase 8.
C (pair core + N-sweep)	$3 \cdot 10^4 \times 12 \times 4 \times 4$	$3 \cdot 10^4 \times 12 \times 4 \times 4$	as registered.
C (degree-2 `nbr` core)	$3 \cdot 10^4 \times 12 \times 4$	$3 \cdot 10^4 \times 12 \times 4$	as registered.
C (attention over complete, pair)	$3 \cdot 10^4 \times 12 \times 7$	$3 \cdot 10^4 \times 12 \times 7$	as registered.
C (attention, degree-2 `nbr`)	$3 \cdot 10^4 \times 12 \times 7$	$3 \cdot 10^4 \times 12 \times 7$	as registered.
R (stabilised GCN, neg. endpoint)	— (post-hoc control)	$1.5 \cdot 10^5 \times 10 \times 2 \times 2$	operator-asymmetry check, App. H.
C (full-obs robustness)	— (post-hoc)	$3 \cdot 10^4 \times 12 \times 4$	robustness cell.
D (TokenMatch replication)	— (post-hoc)	$3 \cdot 10^4 \times 12 \times 4$	out-of-harness replication.

E MPE external validity: full results

This appendix supports the external-validity replication of Section 6.3. Figure 5 is the MPE analogue of the CoordGrid capacity-control figure (Figure 7): at N=3 the three algorithms are indistinguishable, while at N=6 MLP-QMIX pulls above QMIX (capacity helps) and GNN-QMIX drops below both (the graph penalty). The coordination graph is the k=2 nearest-neighbour graph over agent positions, recomputed each episode and held fixed within it (Section 6.3); the trainer reads it once per episode exactly as for CoordGrid.

F External validity on standard benchmarks: LBF and a SMAC check

To test whether the graph penalty is specific to our harness, we ported the controlled contrast into EPyMARL (Papoudakis et al., 2021), a standard cooperative-MARL codebase, and ran it on tasks outside CoordGrid. We add two agent variants to EPyMARL's tuned recurrent QMIX: a *graph* agent that applies inter-agent attention over the per-agent recurrent states (the message-passing operator of DGN/G2ANet) before the Q-head, and a *parameter-matched control* with the identical network but the inter-agent attention fixed to the identity (each agent uses only its own value vector). The two have byte-identical parameter counts, so their gap isolates the graph exactly as GNN-QMIX vs. MLP-QMIX does in the main text. Note this replication uses a *different* relational operator from the main text—inter-agent *attention* over *recurrent*

Table 7: Phase 3 graph-vs-capacity decomposition on MPE `simple_spread` (mean final-window return over 10 seeds; less negative is better). At N=6 the *capacity* effect (MLP-QMIX − QMIX) is positive and the *graph* effect (GNN-QMIX − MLP-QMIX) is sharply negative, so they cancel and the naive GNN-QMIX-vs-QMIX comparison looks null; only the matched control exposes the graph penalty. d is Cohen's d; * significant under Holm and Mann–Whitney.

	QMIX	MLP-QMIX	GNN-QMIX	capacity effect MLP-QMIX − QMIX	graph effect GNN-QMIX − MLP-QMIX
$N = 3$	-44.9	-44.3	-42.4	+0.6 (ns)	+2.0 (ns)
$N = 6$	-74.3	-59.6	-80.4	+14.6 (d= + 1.03)	-20.8^* (d= − 1.46)

Table 8: Full Phase 3 results: final-window return (mean [95% CI] over 10 seeds) for all five algorithms on MPE `simple_spread`. The graph-vs-capacity decomposition (Table 7) concerns the QMIX family (QMIX/MLP-QMIX/GNN-QMIX), which share the monotonic mixer. The mixer-free IQL and VDN baselines lead outright at this budget (15% of the registered steps), which is exactly why Section 6.3 scopes Phase 3's evidence to the within-family decomposition at a fixed budget rather than to converged final performance.

Algorithm	$N = 3$	$N = 6$
IQL	−19.4 [−19.8, −19.0]	−33.2 [−33.5, −32.8]
VDN	−21.5 [−22.9, −20.1]	−34.9 [−36.0, −33.9]
QMIX	−44.9 [−49.4, −40.5]	−74.3 [−85.9, −62.7]
MLP-QMIX	−44.3 [−48.2, −40.4]	−59.6 [−68.2, −51.0]
GNN-QMIX	−42.4 [−50.9, −33.9]	−80.4 [−92.0, −68.9]

states, rather than the feedforward GCN of GNN-QMIX—so it is a *conservative* check: if the penalty appears even with a different, separately-parameterised graph mechanism, the mechanism finding is at least as robust as a same-operator replication would show.

Level-Based Foraging. We run two cooperative LBF tasks (Papoudakis et al., 2021) at 5 seeds and 2×10^6 steps (final-window mean return; Table 9); the pattern tracks team size. On `8x8-2p-2f-coop` (2 agents) no pairwise difference is significant—the two-agent "graph" is a single neighbour and seed variance dominates. On `10x10-3p-3f` (3 agents), GNN-QMIX is *significantly worse* than QMIX ($\Delta = -0.071$, Cohen's $d = -2.24$, Welch $p = 0.014$, Mann–Whitney $p = 0.032$), while MLP-QMIX matches QMIX (capacity neutral). The GNN-QMIX-vs-control contrast points the same way ($d = -1.11$) but is *not* significant at n=5, so we report it as direction-only and do not lean on it. The graph is thus *neutral with two agents and harmful with three*—an independent, standard-benchmark echo of the team-size scaling of the penalty on CoordGrid (Appendix J). *Data provenance:* these runs were executed in the external EPyMARL framework rather than in this paper's harness; Table 9 reports their summary statistics in full, and the raw EPyMARL logs are not part of the anonymized code artifact.

Table 9: LBF: final-window mean return, mean (SD) over 5 seeds at 2×10^6 steps. The graph is neutral at 2 agents and significantly below QMIX at 3 agents ($d = -2.24$, Welch $p = 0.014$, MWU $p = 0.032$).

Map	QMIX	MLP-QMIX (control)	GNN-QMIX (graph)
`8x8-2p-2f-coop` (2 agents)	0.65 (0.23)	0.48 (0.01)	0.66 (0.24)
`10x10-3p-3f` (3 agents)	0.21 (0.03)	0.22 (0.10)	**0.14 (0.03)**

StarCraft II (`3m`), preliminary. As a single-seed check on the benchmark where graph-based methods are most often credited with gains (Samvelyan et al., 2019), we trained the same trio on SMAC `3m` (2×10^6 steps) and evaluated each over 100 test episodes. QMIX and the matched control both reach a 1.00 win rate; the graph agent reaches 0.92. The map is near-saturated and this is a single seed with no variance estimate, so we report it as a *non-inferential anecdote only*—it cannot support "neutral" or "directionally consistent" as evidence, and we do not carry it into the Discussion's evidence chain. A full multi-seed SMAC study is left to future work, as it is GPU-bound beyond our compute budget.

Takeaway. On both standard benchmarks the inter-agent graph is never positive: it shows no significant advantage over a parameter-matched no-graph control anywhere, and sits significantly below QMIX at three agents on LBF. With the unstabilised-graph deficit on CoordGrid and MPE (Section 6.5), the graph's measured effect *in these full-information / convention-reducible settings* ranges from neutral (or, in its unstabilised form, harmful) and is never positive—in direct contrast with Section 7, where the task-matched graph is decisively beneficial once the task forces routing of private information.

G Robustness across graph topologies

The headline capacity control (Section 6.5, Table 19) uses the `ring` graph. To check that the identified graph penalty is not an artefact of that one topology, we repeat the entire control (same five-vs-three algorithms, same 10 seeds, same 150,000-step budget, same locked hyperparameters) on the `line` graph, whose larger diameter is the harder case for a shallow (L=2) GNN to cover. Table 10 reports the result: the capacity contrast (MLP-QMIX − QMIX) is null at both team sizes, while the graph contrast (GNN-QMIX − MLP-QMIX) is sharply negative and grows with N, exactly as on the ring. Both decisive contrasts are significant under Holm correction over the three-contrast family and under the non-parametric Mann–Whitney test ($p = 0.0002$). As on the ring, GNN-QMIX also carries the largest seed-to-seed variance.

Table 10: Capacity control on the `line` topology (mean final-window return over 10 seeds, 150k steps). Parallel to Table 19: MLP-QMIX tracks QMIX (capacity null) while GNN-QMIX falls far below both (graph penalty), reproducing the ring result on a higher-diameter graph.

	QMIX no cap, no graph	MLP-QMIX cap, no graph	GNN-QMIX cap + graph	graph effect GNN-QMIX − MLP-QMIX
$N = 4$	62.4	63.0	52.9	−10.0
$N = 8$	143.0	144.2	78.5	−65.8

H The full-information penalty is an optimisation pathology of the unstabilised GCN, not of graph aggregation

The negative-endpoint phases (1–2, 8, 10, 11) run the *plain* GCN—no residual connection, no LayerNorm—whereas every positive-endpoint arm runs the *stabilised* ("repaired") GCN (§7.1). This appendix asks the question that asymmetry raises directly: does the full-information penalty survive when the negative-endpoint GCN is given the *same* stabilisation as the positive-endpoint one? We rerun GNN-QMIX with residual+LayerNorm on the decisive Phase 8 cells of *both* topologies (CoordGrid `ring` and `line`, $N \in \{4, 8\}$, 10 seeds, 150,000 steps; identical to Tables 19 and 10 otherwise).

Table 11: Stabilised GCN at the negative endpoint (CoordGrid `ring` and `line`, mean final-window return over 10 seeds, 150k steps). Adding residual+LayerNorm recovers GNN-QMIX to *parity* with the no-graph controls on both topologies—the $d \approx -4$ penalty and the $\sim 9\times$ variance inflation of the plain GCN both vanish. The graph is now *neutral*, not harmful; it still confers no benefit.

		QMIX	MLP-QMIX	GNN-QMIX (plain)	GNN-QMIX (stabilised)
ring	$N = 4$	83.0 (sd 1.9)	83.4 (sd 1.4)	72.8 (sd 4.5)	84.1 (sd 0.3)
	$N = 8$	164.9 (sd 9.2)	164.9 (sd 3.3)	78.3 (sd 30.9)	164.3 (sd 7.0)
line	$N = 4$	62.4 (sd 0.8)	63.0 (sd 0.4)	52.9 (sd 7.3)	62.7 (sd 1.5)
	$N = 8$	143.0 (sd 6.4)	144.2 (sd 2.7)	78.5 (sd 29.5)	143.5 (sd 4.0)

Table 11 is unambiguous, and the recovery replicates across both coordination-graph topologies. On the `ring`, the stabilised GCN matches both no-graph controls at both team sizes: at N=8 it reaches 164.3 against MLP-QMIX's 164.9 (Welch $p = 0.79$, Mann–Whitney $p = 0.19$, Cohen's $d = -0.12$) and QMIX's 164.9 ($p = 0.86$); at N=4, 84.1 vs 83.4/83.0 ($p = 0.19/0.10$). On the higher-diameter `line` the picture is identical: 143.5 vs 144.2/143.0 at N=8 ($|d| \leq 0.22$, Welch $p \geq 0.62$, MWU $p \geq 0.85$) and 62.7 vs 63.0/62.4 at N=4 ($|d| \leq 0.29$, $p \geq 0.16$ on all tests). The plain-GCN gaps of Tables 19 and 10 (-86.6 and -65.8 at N=8, $d \approx -4$) collapse to statistical noise, and the seed-standard-deviation inflation (30.9/29.5 plain vs 3.3/2.7 for MLP-QMIX) is gone (7.0/4.0 stabilised). Two conclusions follow, and we fold both into the main text. *(i)* The dramatic full-information harm—the $d \approx -4$ deficit, the $\sim 9\times$ variance, the growth to N=16—is an *optimisation pathology of the standard unstabilised GCN*, not an intrinsic cost of graph aggregation: the same aggregation, stabilised, trains fine on both topologies. *(ii)* Even stabilised, the graph delivers *no benefit* in this full-information regime—it merely matches the no-graph control, never beats it, on every cell tested. The negative endpoint is therefore, precisely stated, "the graph does not help here," with the stronger "actively harmful" reserved for the common unstabilised operator. *Scope:* this recovery is demonstrated on the CoordGrid `ring` and `line` cells; the MPE `simple_spread` and Level-Based Foraging negatives (§6.3, Appendix F) use the plain / standard operator and were not rerun under stabilisation, so we claim the *no-benefit* result—not the specific stabilisation-recovery mechanism—as the cross-environment finding.

I Graph attention: single- and multi-head (GAT-QMIX, DGN-QMIX)

To test whether the graph penalty is specific to the fixed Kipf–Welling GCN aggregation, we add two graph-*attention* variants that share GNN-QMIX's encoder and monotonic mixer but replace the degree-normalized aggregation with learned attention: GAT-QMIX (single-head additive attention; Veličković et al., 2018) and DGN-QMIX (multi-head scaled-dot-product attention, the relational "convolution" of the attention-based graph-MARL methods we discuss, Jiang et al., 2020; Liu et al., 2020). Both carry *more* parameters than the MLP-QMIX control (DGN-QMIX, with per-head Q/K/V/output projections, by 1.5–2.2× depending on the cell's input dimensions: 2.2× at CoordGrid N=4, 2.0× at N=8, 1.5× on MPE N=6), so an attention-below-MLP-QMIX result is conservative. We run both on the decisive cells under the identical protocol (10 seeds, 150,000 steps).

Table 12: Attention variants vs the QMIX family (mean final-window return, 10 seeds). Both single-head (GAT-QMIX) and multi-head dot-product (DGN-QMIX, the DGN/G2ANet operator) attention land above GNN-QMIX, more expressive attention helps *within* graph methods, but below the no-graph control MLP-QMIX on CoordGrid (the graph still hurts). Last two columns: the attention-graph effects vs the matched no-graph control. * Holm- and Mann–Whitney-significant; † Mann–Whitney-significant only. Holm here is over the per-N attention family (the attention-vs-control contrasts of this appendix), not the six-pair family of Section 5.3; the per-contrast verdict CSVs ship with the artifact.

Env	Cell	QMIX	MLP-QMIX	GNN-QMIX	GAT-QMIX	DGN-QMIX	GAT-QMIX−MLP-QMIX	DGN-QMIX−MLP-QMIX
CoordGrid (ring)	$N = 4$	83.0	83.4	72.8	75.6	73.2	−7.8*	−10.3*
	$N = 8$	164.9	164.9	78.3	110.1	131.4	−54.8*	−33.5*
MPE `spread`	$N = 3$	−44.9	−44.3	−42.4	−52.5	−40.2	−8.1†	+4.1
	$N = 6$	−74.3	−59.6	−80.4	−71.9	−68.7	−12.2	−9.1

J Scaling the graph penalty to larger teams

Finding 5 (Section 6.5) reports that the graph effect grows with team size. The full scaling sweep adds CoordGrid `ring` at $N \in \{12, 16\}$ for QMIX, MLP-QMIX, and GNN-QMIX (10 seeds, 150,000 steps), combined with the $N \in \{4, 8\}$ cells from Table 19. Figure 10 and Table 13 give the result.

Table 13: Larger-N scaling on CoordGrid `ring` (mean final-window return, 10 seeds). The graph effect GNN-QMIX − MLP-QMIX grows monotonically; the no-graph MLP-QMIX control stays within a few points of QMIX at every N, so essentially the entire (and growing) gap is the graph. Both *graph* contrasts are significant at every N under Holm and Mann–Whitney. The *capacity* contrast (MLP-QMIX − QMIX) is not always exactly null: at N=12 it is −7.1 (significant under both tests) and at N=16 −8.6 (Mann–Whitney only)—about 3% of QMIX's return and an order of magnitude smaller than the graph effect at those scales.

	QMIX	MLP-QMIX	GNN-QMIX	graph effect (GNN-QMIX − MLP-QMIX)
$N = 4$	83.0	83.4	72.8	−10.6
$N = 8$	164.9	164.9	78.3	−86.6
$N = 12$	252.6	245.5	92.1	−153.4
$N = 16$	336.6	328.0	60.6	−267.4

K Full per-condition tables

The main text reports Phase 2 via the forest plot and per-condition curves (Figures 3, 4) and the hypothesis tests. The Phase 1 pilot curves, the per-algorithm Phase 1 summary, the full Phase 2 per-condition returns, and the Phase 1 pairwise sanity check are collected here.

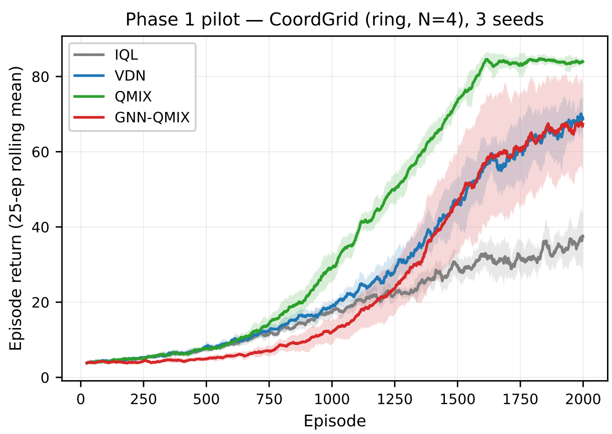

Figure 15: Phase 1 pilot learning curves on CoordGrid ($N = 4$, ring), ten seeds. Shaded bands are ±1 s.d. across seeds of the 25-episode rolling mean. The coordinated baselines (VDN, QMIX) clearly separate from IQL, and GNN-QMIX runs to completion, the Phase-0 harness sanity check (Section 6.1).

Table 14: Phase 1 final-window mean return ($N = 4$, ring, 10 seeds). Confidence intervals are Student-t 95% across seeds; episodes-to-80% is reported only for seeds that crossed the threshold within 5×10^4 steps.

Algorithm	n_{seeds}	Final mean	95% CI	Episodes to 80%
IQL	10	34.62	[32.61, 36.64]	1375
VDN	10	66.023	[62.57, 69.48]	1528
QMIX	10	84.015	[83.41, 84.62]	1431
GNN-QMIX	10	66.015	[57.11, 74.92]	1552

L Supplementary figures

This appendix collects the per-condition depth–diameter curves that accompany the Phase 4 heatmap (Figure 6) in the main text.

Table 15: Phase 2 final-window mean return by condition. Edge density is matched between `ring` and `erdos_renyi` by construction.

graph	N	algorithm	Final mean	95% CI	Episodes to 80%
erdos_renyi	4	GNN-QMIX	7.05	[6.29, 7.81]	427
erdos_renyi	4	IQL	34.05	[31.99, 36.10]	785
erdos_renyi	4	QMIX	82.52	[80.35, 84.70]	873
erdos_renyi	4	VDN	57.32	[54.59, 60.05]	914
erdos_renyi	8	GNN-QMIX	9.18	[8.37, 9.98]	40
erdos_renyi	8	IQL	43.26	[40.47, 46.05]	795
erdos_renyi	8	QMIX	108.79	[82.73, 134.84]	961
erdos_renyi	8	VDN	69.19	[53.99, 84.39]	969
ring	4	GNN-QMIX	34.76	[24.96, 44.56]	974
ring	4	IQL	28.86	[27.10, 30.61]	745
ring	4	QMIX	83.29	[81.43, 85.15]	876
ring	4	VDN	52.64	[46.74, 58.54]	919
ring	8	GNN-QMIX	19.52	[13.88, 25.17]	716
ring	8	IQL	40.10	[34.47, 45.73]	787
ring	8	QMIX	129.13	[100.90, 157.36]	938
ring	8	VDN	70.53	[59.12, 81.94]	908

Table 16: Phase 1 sanity check: IQL vs. each coordinated baseline on final-window mean return ($N = 4$, ring, 10 seeds). Welch's t, Holm-corrected over the 6-pair family.

Comparison	Δ mean return	p_{raw}	p_{Holm}
IQL vs. VDN	31.40	< 0.001	< 0.001
IQL vs. QMIX	49.39	< 0.001	< 0.001
IQL vs. GNN-QMIX	31.39	< 0.001	< 0.001

Table 17: H1 (within-condition): GNN-QMIX minus QMIX in mean final-window return. p-values are one-sided Welch t (H_a: GNN-QMIX > QMIX), not corrected (single test per condition). The ordering is negative in every cell.

N	graph	Δ = GNN-QMIX − QMIX	p (1-sided)
4	erdos_renyi	-75.47	1.000
4	ring	-48.54	1.000
8	erdos_renyi	-99.61	1.000
8	ring	-109.61	1.000

Table 18: H2 (difference-of-differences): the GNN-QMIX vs. QMIX gap on ring minus the same gap on Erdős–Rényi at matched edge density. One-sample t on per-seed DiD against 0 (H_a: DiD > 0).

N	Δ_{ring}	Δ_{ER}	DiD	p (1-sided)
4	-48.54	-75.47	26.93	< 0.001
8	-109.61	-99.61	-10.00	0.701

M Reproducibility checklist

Select the answers that apply to this research – one per item.

All articles:

1. All claims investigated in this work are clearly stated. **[yes]**

Table 19: Phase 8 final-window return (mean over 10 seeds) on the `ring` graph (the numbers behind Finding 1, Section 6.5). MLP-QMIX is the parameter-matched no-graph control (identical parameter count to GNN-QMIX). *Capacity* effect = MLP-QMIX − QMIX; *graph* effect = GNN-QMIX − MLP-QMIX.

	QMIX no cap, no graph	MLP-QMIX cap, no graph	GNN-QMIX cap + graph	graph effect GNN-QMIX − MLP-QMIX
$N = 4$	83.0	83.4	72.8	-10.6 ($d = -3.2$)
$N = 8$	164.9	164.9	78.3	-86.6 ($d = -3.9$)

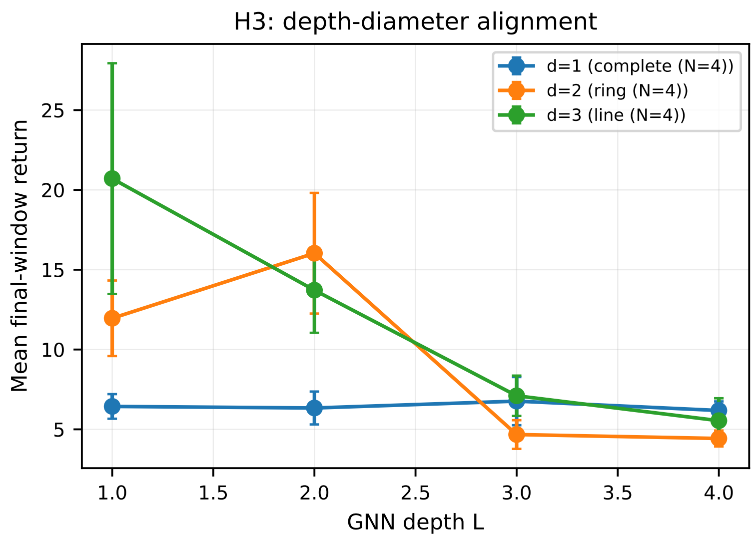

Figure 16: Phase 4 depth–diameter curves (companion to the heatmap in Section 6.4). Final-window return vs. GCN depth L for each graph diameter d; the over-depth cliff at $L \geq 3$ is visible on the structured diameters ($d = 2, 3$), while the complete graph ($d = 1$) is flat in L.

2. Clear explanations are given how the work reported substantiates the claims. [**yes**]
3. Limitations or technical assumptions are stated clearly and explicitly. [**yes**]
4. Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. [**yes**]
5. Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. [**yes**]

Articles containing theoretical contributions:

Does this paper make theoretical contributions? [**no**]

Articles reporting on computational experiments:

Does this paper include computational experiments? [**yes**]

1. All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for source code readability and documentation as well as for long-term accessibility. [**yes**]
2. The source code comes with a license that allows free usage for reproducibility purposes. [**yes**]

3. The source code comes with a license that allows free usage for research purposes in general. [**yes**]
4. Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility. [**yes**]
5. The unaggregated data comes with a license that allows free usage for reproducibility purposes. [**yes**]
6. The unaggregated data comes with a license that allows free usage for research purposes in general. [**yes**]
7. If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. [**yes**]
8. The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models; amount of memory (cache and RAM); make and version of operating system; names and versions of relevant software libraries and frameworks. [**yes**]
9. The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. [**yes**]
10. The number of algorithm runs used to compute each result is reported. [**yes**]
11. Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. [**yes**]
12. Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. [**yes**]
13. All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. [**yes**]
14. The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation. [**partially**]
15. Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. [**yes**]

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? [**yes**]

1. All newly introduced data sets are included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility with a license that allows free usage for research purposes. [**yes**]
2. The newly introduced data set comes with a license that allows free usage for reproducibility purposes. [**yes**]
3. The newly introduced data set comes with a license that allows free usage for research purposes in general. [**yes**]
4. All data sets drawn from the literature or other public sources (potentially including authors' own previously published work) are accompanied by appropriate citations. [**yes**]
5. All data sets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. [**yes**]

6. All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant. [**yes**]

7. All methods used for preprocessing, augmenting, batching or splitting data sets (e.g., in the context of hold-out or cross-validation) are described in detail. [**NA**]

Explanations on any of the answers above (optional):

All environments (the CoordGrid gridworld and the TokenMatch referential game) are procedural generators released with the code (MIT-licensed); the external MPE and LBF tasks are cited and publicly available. Hyper-parameters were pre-registered and locked before confirmatory runs, so per-experiment tuning effort was deliberately minimal. On raw data: the supplementary artifact ships every aggregate and per-contrast statistics CSV plus the exact per-run configuration (seed, hyperparameters, budget) for every run; per-episode logs are regenerated bit-identically from those configurations (training is fully seeded and deterministic on CPU) and will be published with the de-anonymized repository. The Level-Based Foraging and SMAC checks (Appendix F) were executed in the external EPyMARL framework; their summary statistics are reported in full in Table 9, and their raw logs live outside this artifact. On unsuccessful experiments: the search that preceded the positive endpoint, including two constructions that failed to produce a significant graph advantage, is disclosed in Section 7.1 ("How we found this task").